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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

Team member 1	**Bianka Toneva**
Team member 2	**Dipanshu Sharma**
Team member 3	**Miklos Lammel**

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.
Note: You may be required to provide proof of your outreach to non-contributing members upon request.

(N/A)

Note: Please refer to the Python notebook for executable code, extensive commentary, and results for each question in this document.

Step 1 - Individually reading the paper

Everyone has read the paper by Alvi (2018).

Step 2 - Problem formulation

Crude oil prices have a significant impact on global industries. Forecasting crude oil prices has been a critical and enduring challenge. In the author's dissertation, '*Application of Probabilistic Graphical Models in Forecasting Crude Oil Price*', Danish A. Alvi addresses this complex problem by leveraging Probabilistic Graphical Models (PGMs), specifically Bayesian Networks.

The paper focuses particularly on the interactions between geopolitical risk, uncertainty within the market, and other indicators such as GDP and inflation. This paper aims to integrate multiple dimensions of risk and uncertainty into a unified framework so that it can address the need for more robust forecasting models, which will in turn allow investors to be more risk-averse.

The GPR quantifies the level of geopolitical tension, which is a key driver of oil price volatility. The EUI measures uncertainty in the energy sector, providing insights into potential supply disruptions or demand fluctuations. Both indices, being monthly and serving as proxies for broader political and energy-specific risk, are crucial for robust oil price forecasting.

The author primarily focuses on extensive data cleaning, including detecting missing values or clearing ”NaN”. Outliers are then removed by using the IQR-based formula, and the data is merged into a monthly time index. During the exploratory data analysis, which follows, the author decides to explore Bayesian Networks in order to highlight how direct and indirect factors can affect the crude oil price, offering a framework for simulating how a shock to one variable, like a sudden geopolitical crisis, might propagate through the oil market and broader economy.

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What is the problem the student's thesis attempts to solve?

The global crude oil market is renowned for its extreme volatility and non-linearity. It is influenced by hundreds, if not thousands, of independent variables, making it a formidable task to predict using traditional time series forecasting models such as GARCH.

Crude oil prices are generally influenced by geopolitical events such as wars and sanctions, recessions, OPEC policies, market speculation, and even natural disasters.

The student has undertaken the ambitious task of creating an accurate predictor of crude oil price, taking into account some of the complex dependencies mentioned. This task requires a high level of determination and commitment, which the student has demonstrated throughout the research process.

To achieve this, the student has identified the complex interactions and dependencies of the different factors affecting crude oil prices. He has created a network where each factor is labelled as a node. Crucially, he has chosen to use a Bayesian Network to model the volatility of the crude oil market, a method well-suited to this problem.

Why are Bayesian networks well-suited to solve it?

Bayesian Networks (BNs) are a type of Probabilistic Graphical Model that represents a set of variables and their conditional dependencies through a Directed Acyclic Graph (DAG). They are well suited for Alvi's problem for several reasons:

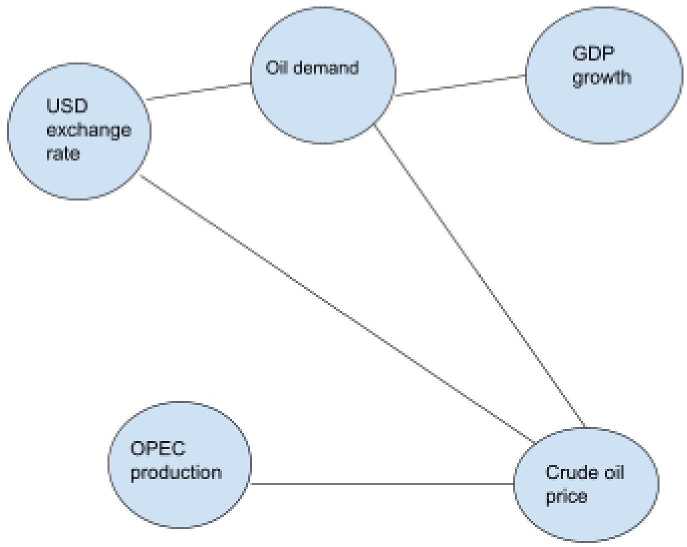
Firstly, it can tackle uncertainties since probability distributions are used. This further allows us to quantify uncertainty in both an input (e.g., macroeconomic forecasts) and an output (price predictions). For crude oil markets in particular, it is common to see shocks that can now be mitigated via probabilities in our model. As previously mentioned, Bayesian networks allow us to model the causal relationships between a series of events that can affect crude oil prices. This is achieved via nodes and edges signifying causal influence.

Bayesian Networks support both diagnostic and predictive inference. They are based on Bayesian probability, which allows us to predict the outcome of event A given that event B has already occurred. We are able to analyze various scenarios and outcomes based on a previously determined event. For example, we could examine the consequences of the fluctuation of crude oil prices, given that a war has broken out in the Middle East.

What are the advantages of using this methodology for this problem?

There are a few main advantages of this method. The DAG structure makes the model's assumptions explicit and visually interpretable, which in turn can allow us to follow and directly observe which factors affect the price and how. The pattern of events is clearer, which highlights key events that trigger fluctuations.

An example of a Bayesian Network that we can visualize is provided below.



Bayesian Networks are able to handle datasets that can be incomplete and do so by estimating the missing values through marginalization.

Furthermore, due to the model's output in the form of probability distributions for different forecasts, analysts are able to align the risk that the investor is willing to take with the probability of a certain price being achieved, given all sorts of prerequisites.

Finally, as mentioned previously, the model allows for analysis of the form "find the probability of event A given that event B took place", meaning the probability of a certain outcome for a vast majority of combinations of events can be explicitly calculated. This leads to better risk mitigation and an in-depth view of probable price fluctuations.

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Step 3 - Macroeconomic, microeconomic, and geopolitical variables

For the GWP1 assignment, we have decided to use the data sources below. For a longer explanation about the used data sources, see Step 4.

Financial Market data

- WTI Crude Price: Benchmark for global oil prices
- US Dollar Index (DXY)
- Volatility Index (VIX)
- Additional datasets for the future:
    - Brent Oil was considered, but not used, because the time series only starts in 2007 on yfinance.

Geopolitical data

- Geopolitical Risk Index (GPR): Captures instability, sanctions, wars, etc., which strongly affect oil
- Energy Uncertainty Index (EUI)
- Additional datasets for future investigation:
    - Economic Policy Uncertainty (EPU) Geopolitical Broader than GPR — includes political and economic instability.
    - US Sanctions Index (OFAC proxy) Geopolitical Sanctions tracker (indirect) from political data or BDS project
    - Trade Policy Uncertainty (TPU) Measures uncertainty in trade-related new

Macroeconomic data

- US GDP Growth Rate: Indicates economic activity & oil demand
- US Consumer Price Index (CPI): Captures inflation; affects oil prices via monetary policy
- Additional datasets for future improvements:
    - Global Oil Demand (OECD), we tried to pull the dataset, but it was not available anymore
    - Global Oil Supply (OPEC), we tried to pull the dataset, but it was not available anymore
    - Interest Rates (10Y)

Microeconomic data

- Refinery Utilization Rate (RUR): This index acts as a proxy for downstream oil demand.
- Weekly Crude Oil Ending Stock US (WCESTUS1) - Proxy for demand and supply

Step 4 - Dictionary and table of the datasets used

The table below summarizes all the various datasets we utilized for our assignment.

Type	**Abbr.**	**Freq.**	**Source**	**From**	**Til**	**Description**
Geopolitical Data	GPR	Monthly	policyuncertainty.com	Jan 1985	June 2025	The Geopolitical Risk Index (GPR) is a measure of adverse geopolitical events and the associated risks based on news coverage.
Geopolitical Data	EUI	Monthly	policyuncertainty.com	Jan 1996	Nov 2023	The Energy Uncertainty Risk Index (EUI) shows the uncertainty in energy markets driven by geopolitical and economic events.
Financial Market Data	WTI	Daily (resampled Monthly)	Yahoo Finance	Jan 2000	Jun 2025	West Texas Intermediate (WTI) is a grade of crude oil and one of the three main benchmarks in oil pricing.
Financial Market Data	VIX	Daily (resampled Monthly)	Yahoo Finance	Jan 1996	Nov 2023	The Volatility Index (VIX) shows the market expectations of near-term volatility, also called the "fear gauge."
Financial Market Data	DXY	Daily (resampled Monthly)	Yahoo Finance	Jan 1996	Nov 2023	The US Dollar Index (DXY) indicates the strength of the US dollar against a basket of major world currencies. A stronger dollar typically suppresses oil prices.
Macroeconomic Data	GDP	Monthly	FRED	Jan 2000	Jun 2025	Real US Gross Domestic Product (GDP) growth is an indicator of overall economic activity and health.
Macroeconomic Data	CPI	Monthly	FRED	Jan 2000	Jun 2025	This is the US Consumer Price Index (CPI) data, showing changes in the price level of a

						basket of goods. Acts as a proxy for inflation.
Microeconomic Data	RUR	Monthly	FRED	Jan 2000	Jun 2025	Refinery Utilization Rate (RUR) - the percentage of total refinery capacity in use. Acts as a proxy for downstream oil demand.
Microeconomic Data	WCE STUS	Weekly (resampled Monthly)	EIA	Aug 1982	Jun 2025	Weekly Crude Oil Ending Stock for US - acts as a proxy for downstream oil demand.

Geopolitical Data

GPR (Geopolitical Risk Index)
GPR is a monthly index that measures the intensity of geopolitical tensions based on newspaper coverage. Higher values indicate instability in the world, which then often increases oil price volatility due to intense supply and demand fluctuations. Caldara et al. (2021)

EUI (Energy Uncertainty Index)
This index captures uncertainty in the energy sector linked to geopolitical or economic turmoil. It can help to identify when the oil market unpredictability increases, which can increase speculative price changes. Dang et al. (2023)

Financial Market Data

WTI (West Texas Intermediate)
WTI is a key Benchmark for U.S. crude oil pricing. It shows the spot price of oil, which is directly influenced by global supply-demand conditions. Along with Brent and Dubai Crude, TWI is an essential reference point for the oil market analysis.

VIX (Volatility Index)
VIX is the volatility index of the S&P 500, and it measures market expectations of near-term volatility. It serves as a proxy for investor risk appetite, hence the nickname “fear gauge”. Higher VIX can trigger market participants to sell off during market turmoil.

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DXY (U.S. Dollar Index)
The U.S. Dollar Index shows the strength of the U.S. dollar compared to major world currencies. Since oil is priced in dollars, a stronger DXY typically makes oil more expensive for non-US countries to buy globally, which then reduces the demand and thus affects oil prices downward.

Macroeconomic Data

GDP (Gross Domestic Product)
Shows the overall health and activity of the U.S. economy. Strong GDP growth signals higher industrial and consumer energy use, and results usually in boosting the oil demand (among other things). Higher demand naturally increases the oil prices.

CPI (Consumer Price Index)
This index is a measure of inflation based on household goods and services. A bucket is formed for these goods and services, and the changes in the pricing of this bucket are the CPI.
Rising CPI indicates inflation, and the increase in interest rates by the Fed usually counters rising inflation. This monetary tightening usually impacts oil demand, and thus oil prices as well.

Microeconomic Data

RUR (Refinery Utilization Rate)
Indicates the percentage of U.S. refinery capacity currently in use. A high RUR reflects strong downstream demand for crude oil. Higher demand supports the oil price levels.

WCESTUS1 (U.S. Ending Crude Oil Stocks, excl. SPR)
Represents weekly crude inventories (excluding the Strategic Petroleum Reserve), which is a balance of supply and demand. High stock levels can imply oversupply, meaning the demand is lower than the supply, and this creates a downward pressure on oil prices.

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Step 5 - Cleaning the data

Extreme Outliers
Outliers were identified using the Interquartile Range (IQR) method. For each variable, the values outside of the 3xIQR were flagged as extreme. The outliers were reviewed and either removed or retained based on context.

Bad Data
Here, we examined the data for incorrect or illogical values, such as negative inventory levels or more than 100 percent, where the highest achievable proportion is 100 percent. Numeric columns were checked for NaN values.

Missing Values
Missing values were replaced where needed. Linear interpolation was used for short gaps.

Outcome
All datasets are clean, aligned, and regularly sampled time series with consistent date indices and no gaps.

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Step 6 - Why certain data points were eliminated

WTI datapoint eliminated

There was a short period during the 2020 COVID pandemic market crash when the Oil prices dropped below 0 for a day. This caused all sorts of problems in the calculations; thus, this value was eliminated by replacing it with the mean of the previous and following days.

Periods eliminated

Since the various data sources rarely have perfectly overlapping periods, we eliminated those parts where the values were missing for the other time series. As an example, the yfinance database has values for the WTI time series that only started in 2000. So, even though other data sources might be available from an earlier date, we have decided to use only the overlapping portion.

Bad data and outliers eliminated

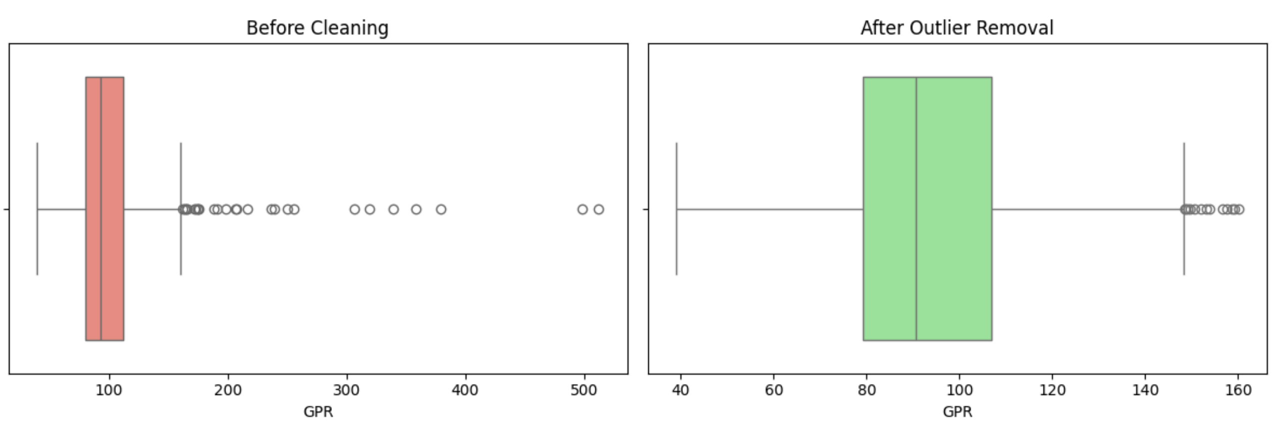
As mentioned above, a few instances of bad data and outliers were eliminated as well. Since some of the sources were from Excel sheets downloaded from the internet, they could contain typos and other mistakes.

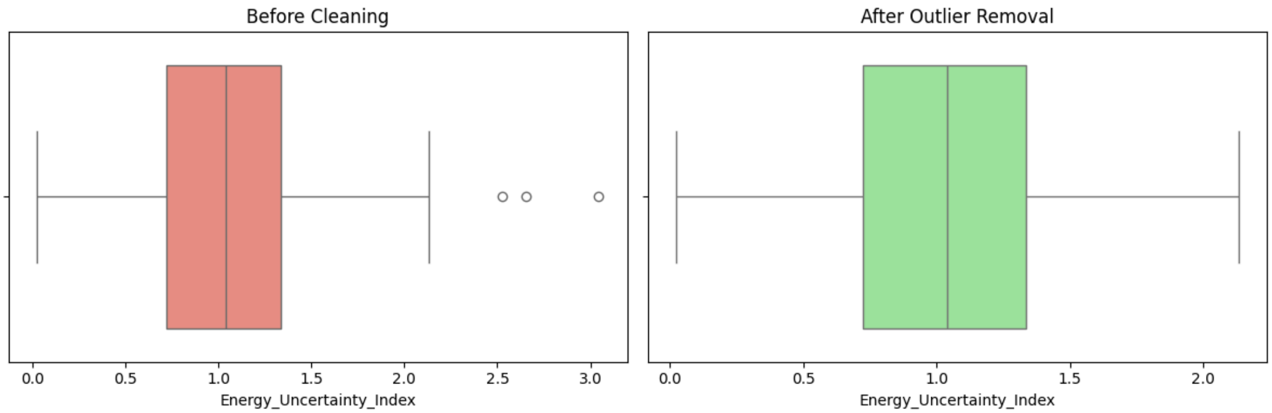
The overall method

First, we started by cleaning the dataset of any outliers by using the IQR method. Then we moved on to replacing missing values by the mean of the series, where necessary.

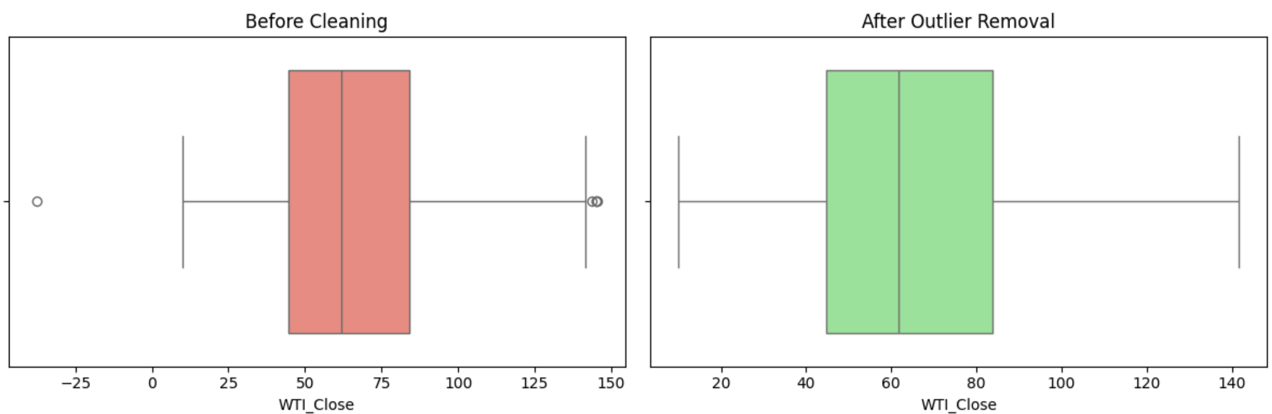
The IQR method involves removing any outliers that are more than 1.5 times the interquartile range away from the first and third quartiles, respectively.

This data cleaning produced the following plots for the GPR and EUI datasets.

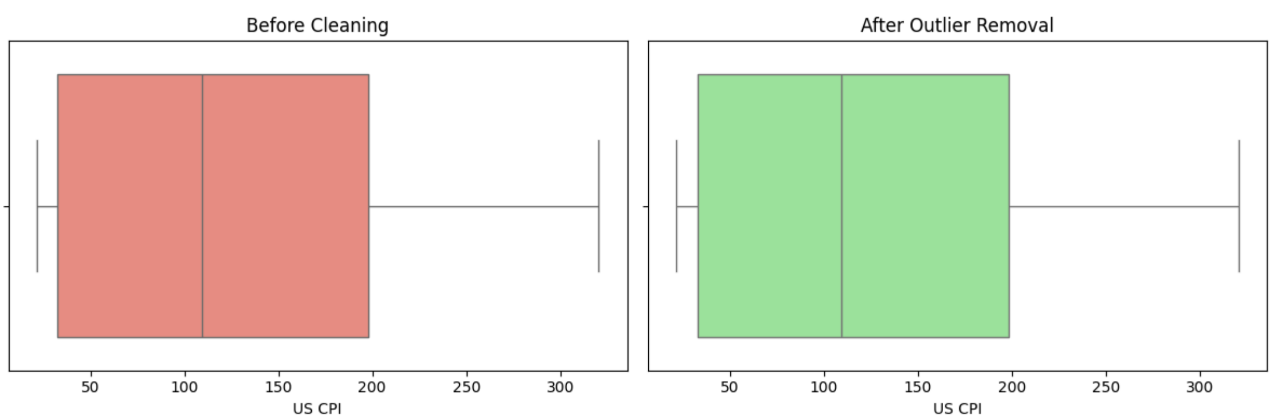




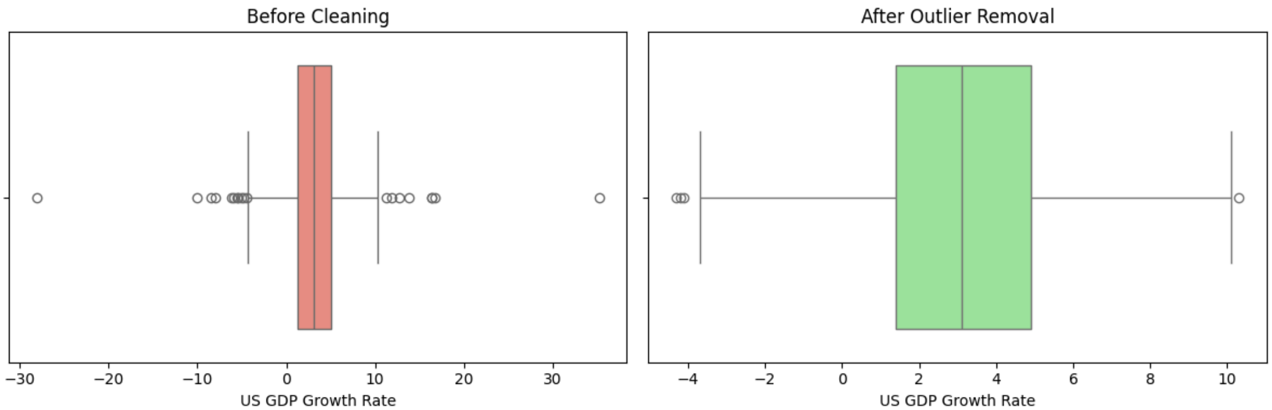
We then repeated the process for the WTI Crude Oil data, with the only additional step being that we only considered prices from $50 to $200 per barrel in order to consider only 'realistic' prices. Our results are visualized in the following box plot:



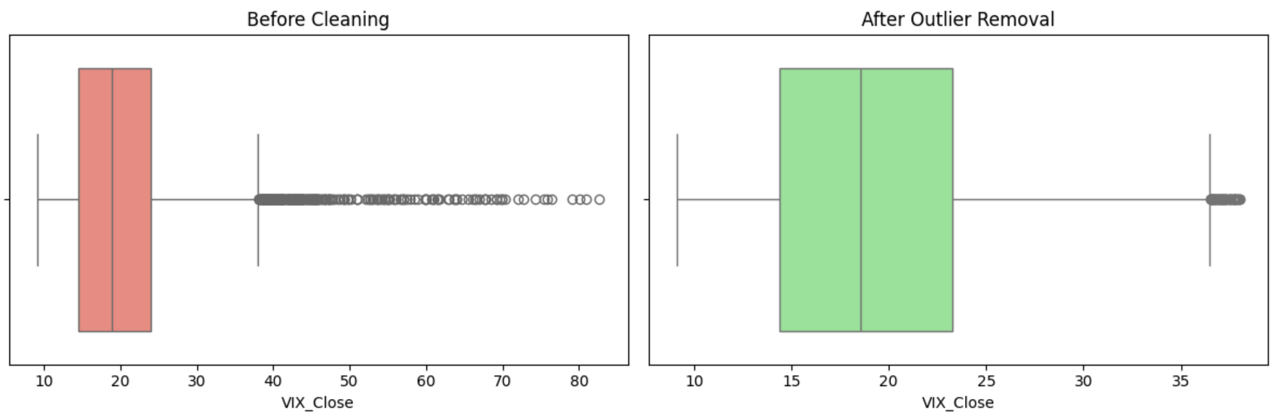
The same method was used for all other datasets. For the US CPI data, we ensured that all values were positive.



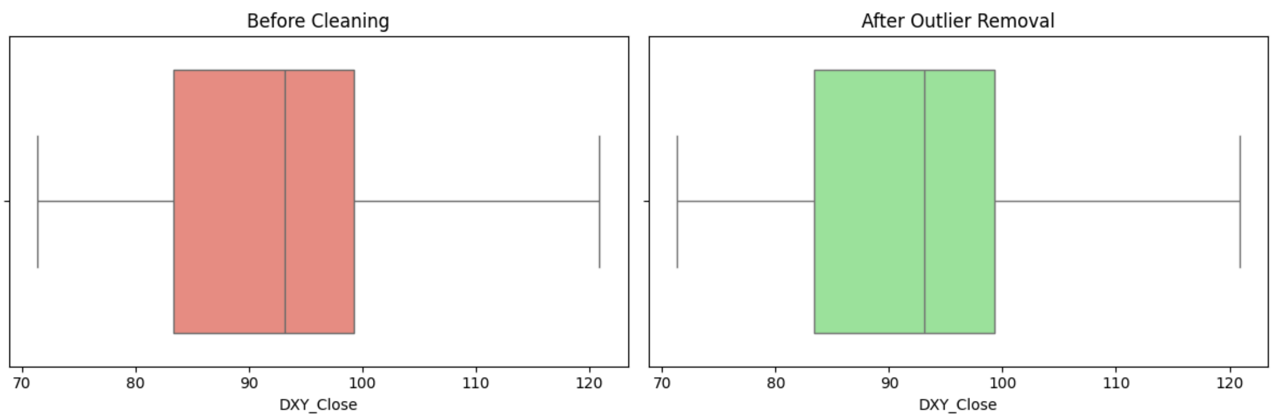
For the US GDP data, a plausible growth rate would be between -50% to 50%.



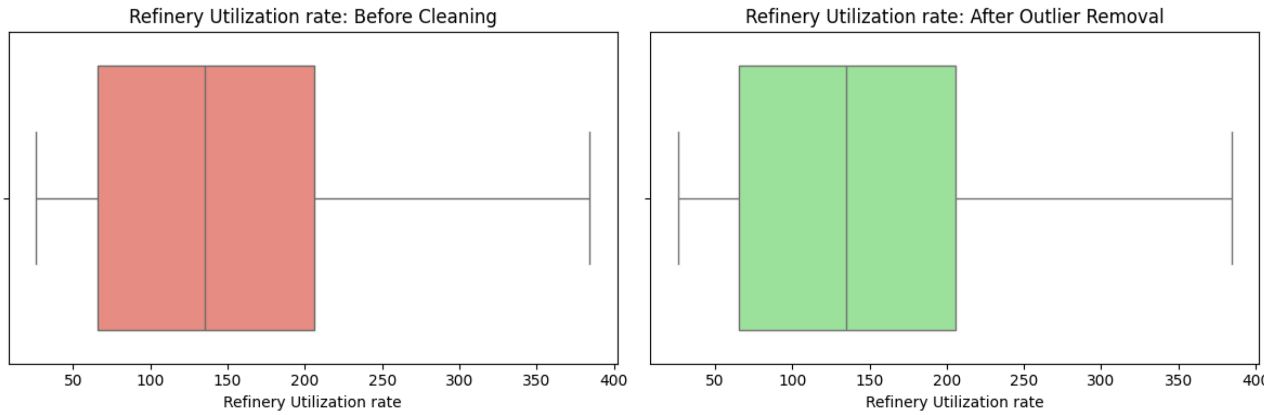
For the VIX Volatility index, we only considered values in the range 0-100.



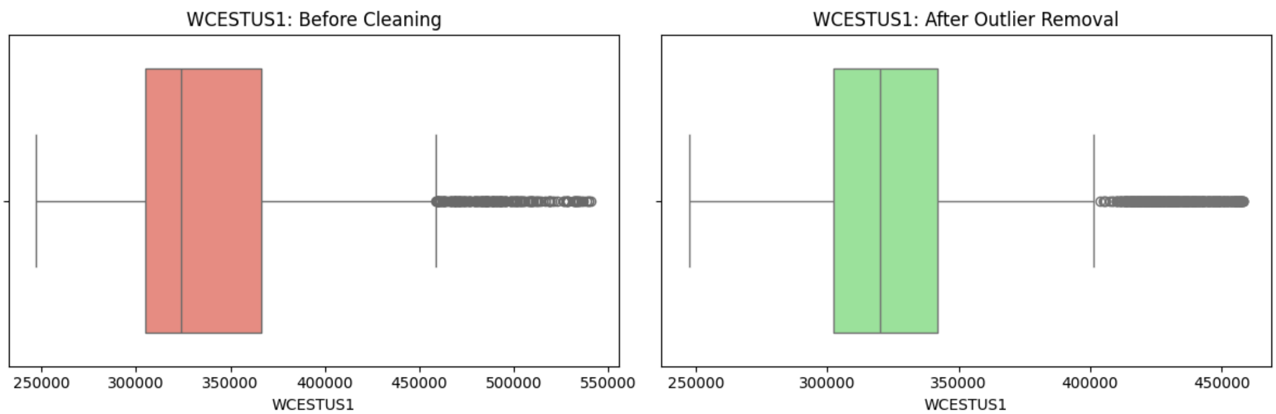
For the DXY US Dollar index, we only considered values from 50 to 150.



For the Refinery Utilization Rate, we made sure that all values were positive.



Finally, for WCESTUS1, we made sure that all values were positive.



Step 7 - Exploratory Data Analysis

For the selected series (GPR (Geopolitical Risk Index), EUI (Energy Uncertainty Risk Index), CPI (Consumer Price Index, Inflation), GDP (Gross Domestic Product), WTI (Oil Price for West Texas Intermediate), and RUR (Refinery Utilization Rate), we have made the Histograms + KDE under distributional plots. We have made line plots for time series plots. We have also plotted a correlation heatmap, pairplot, joint distribution with WTI, and autocorrelation plots under multivariate plots. Below are the results-

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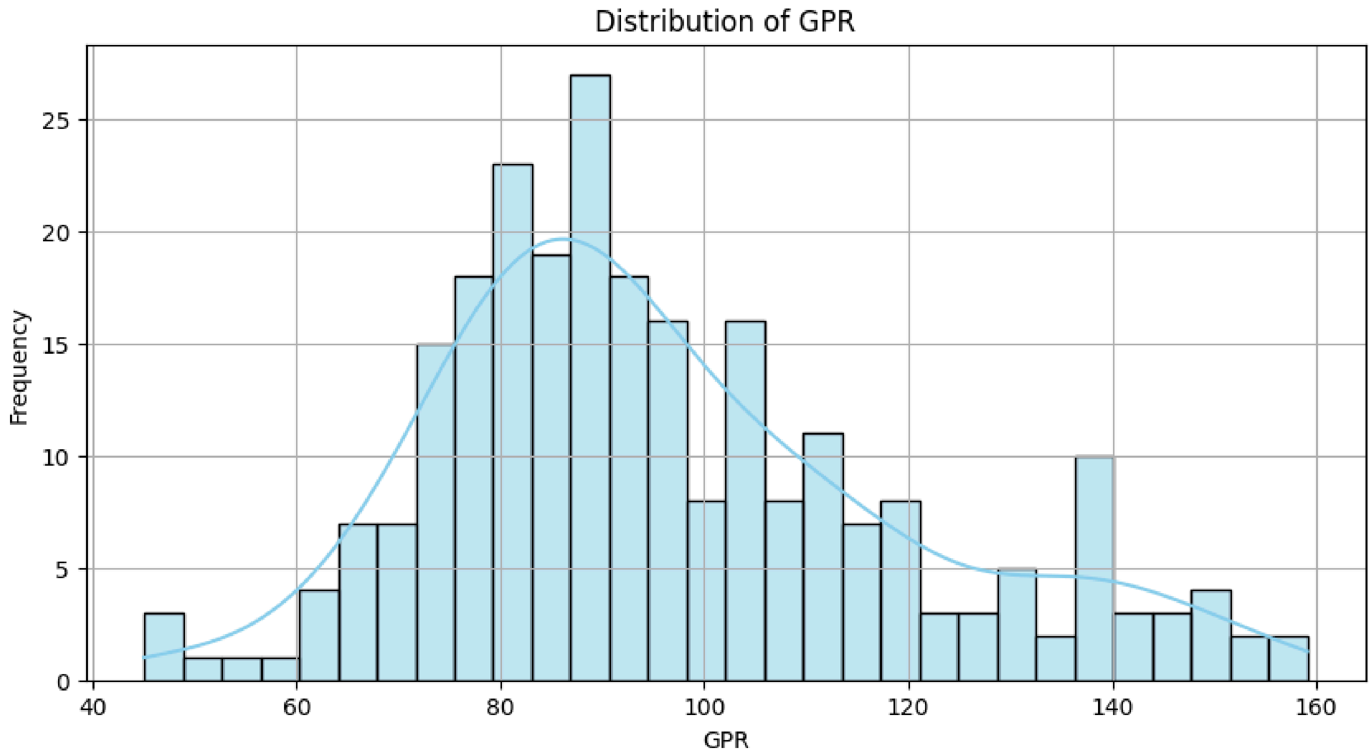
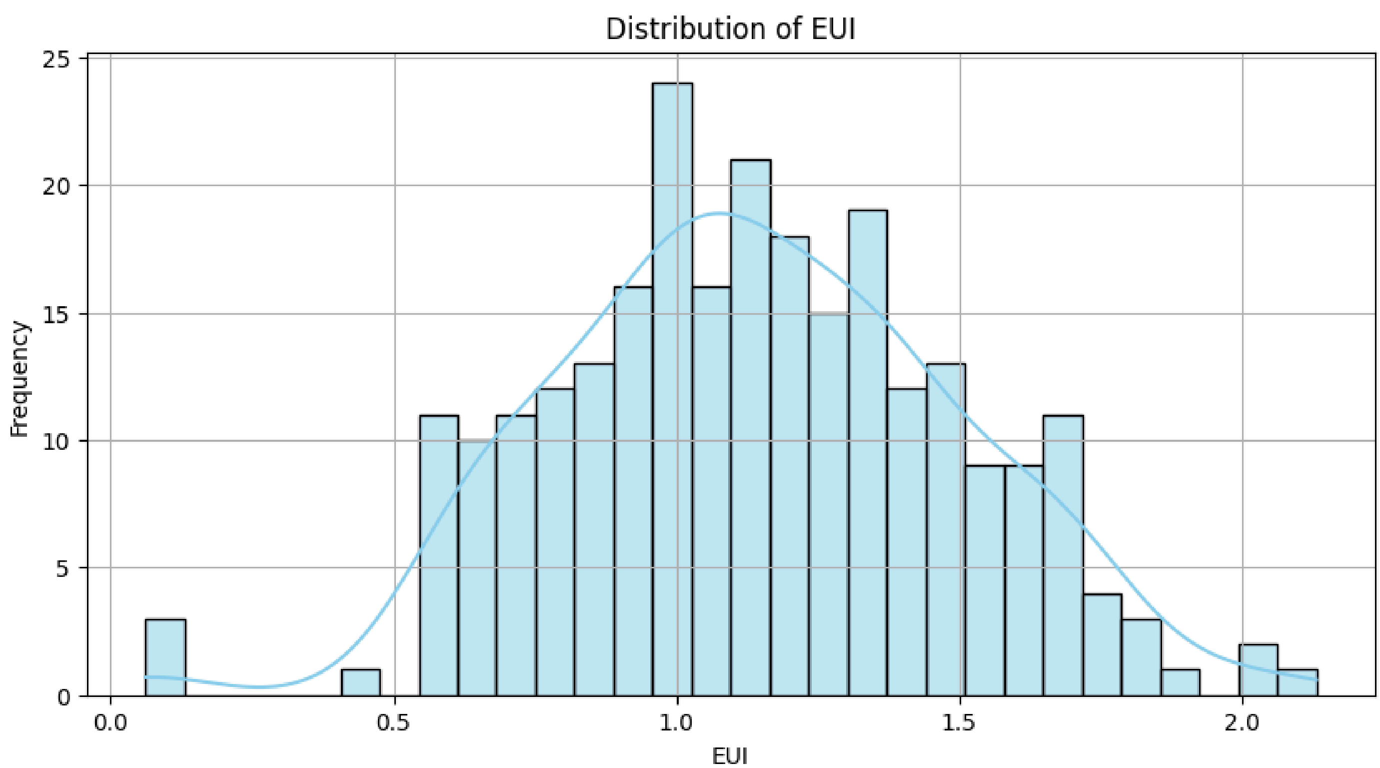
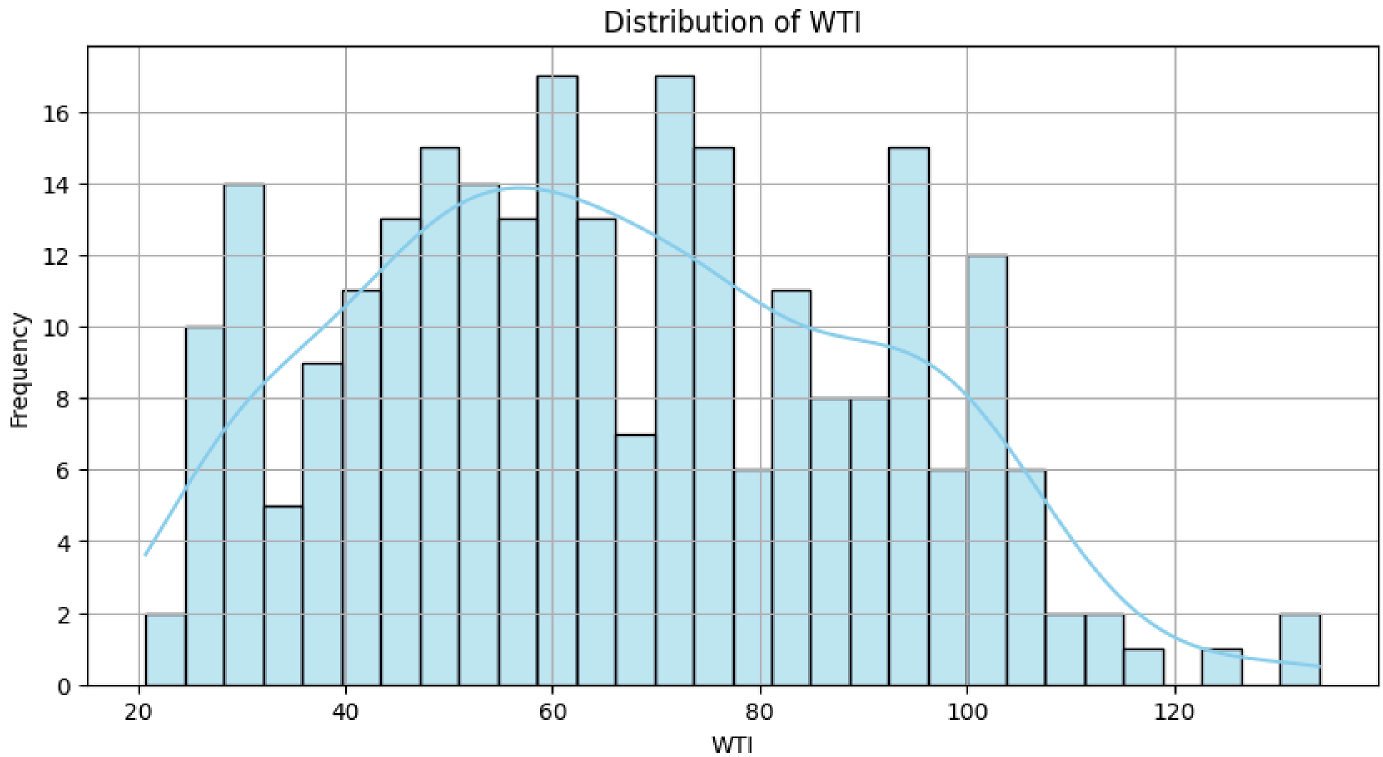
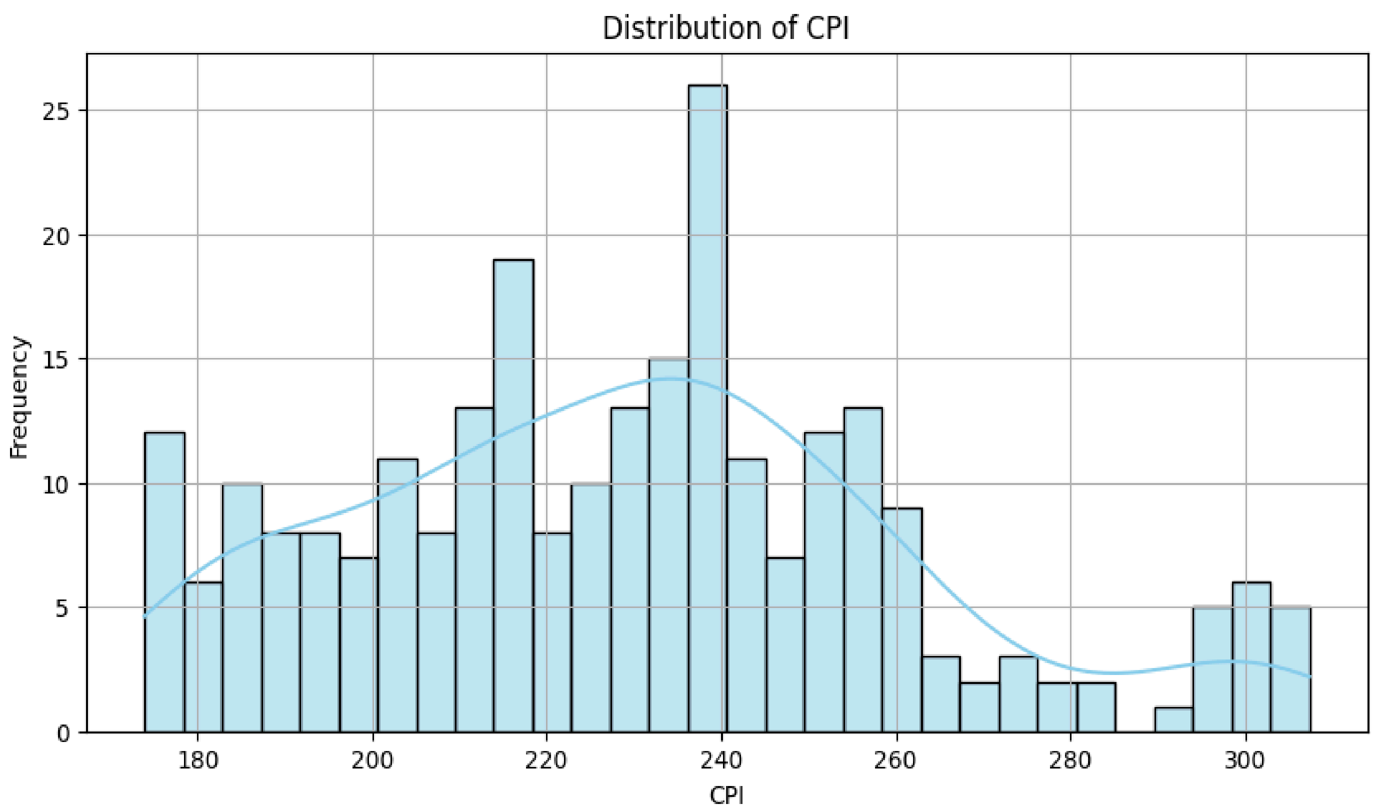
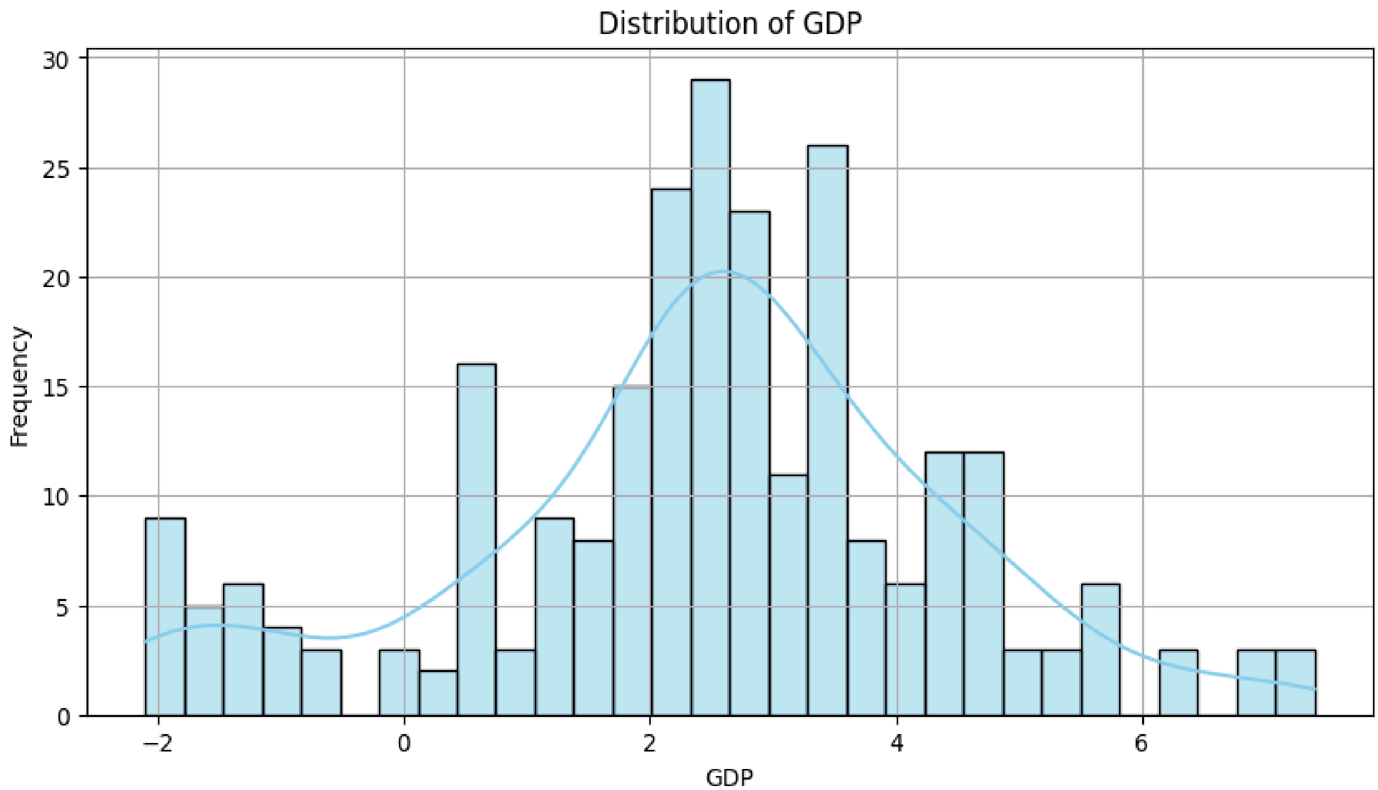
Distributional Plots

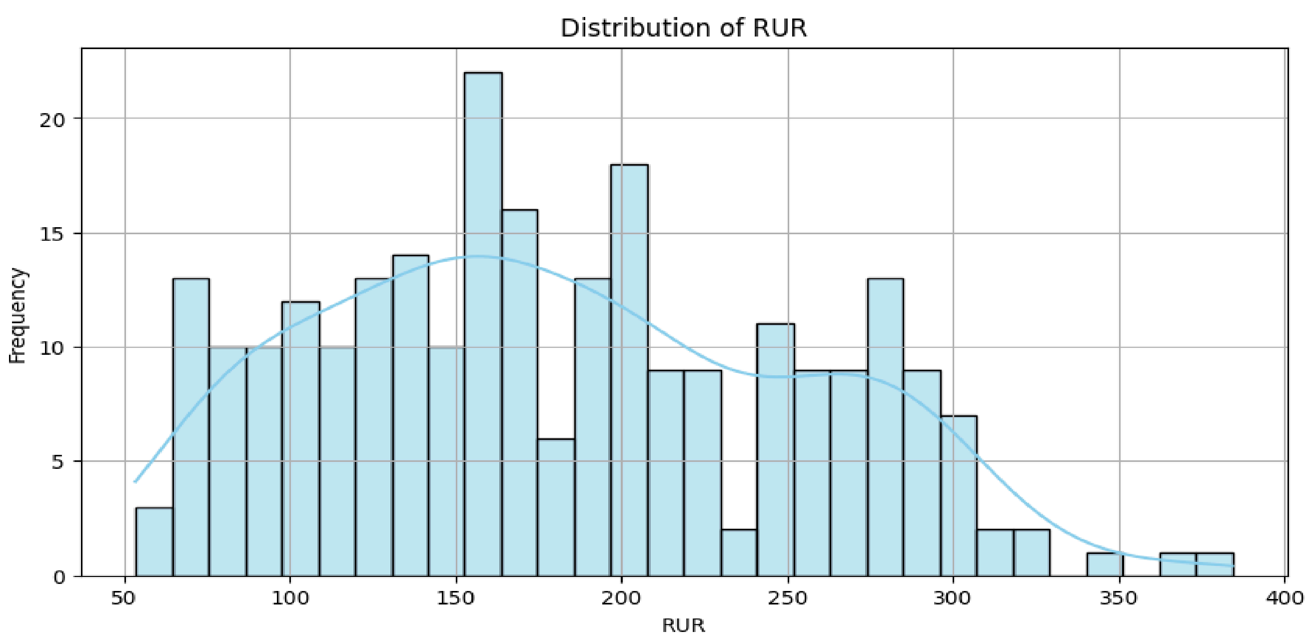


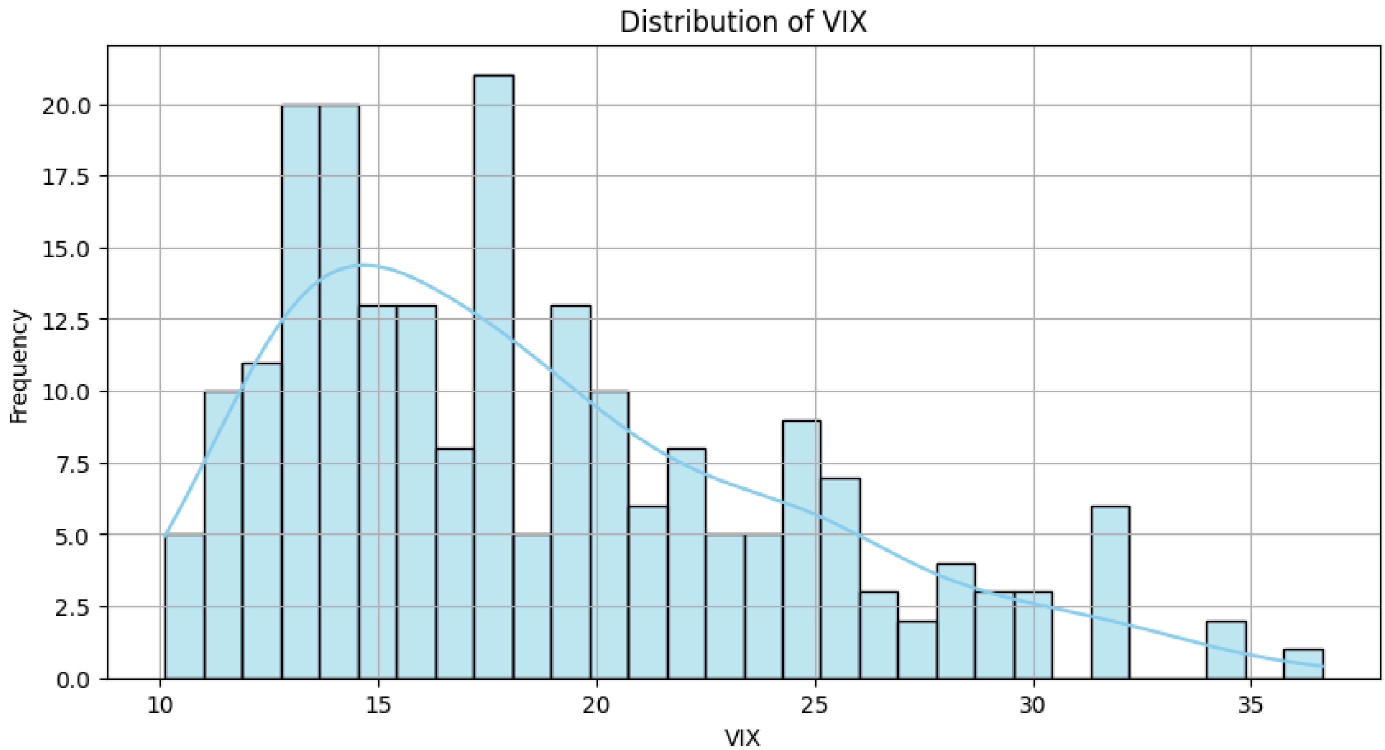


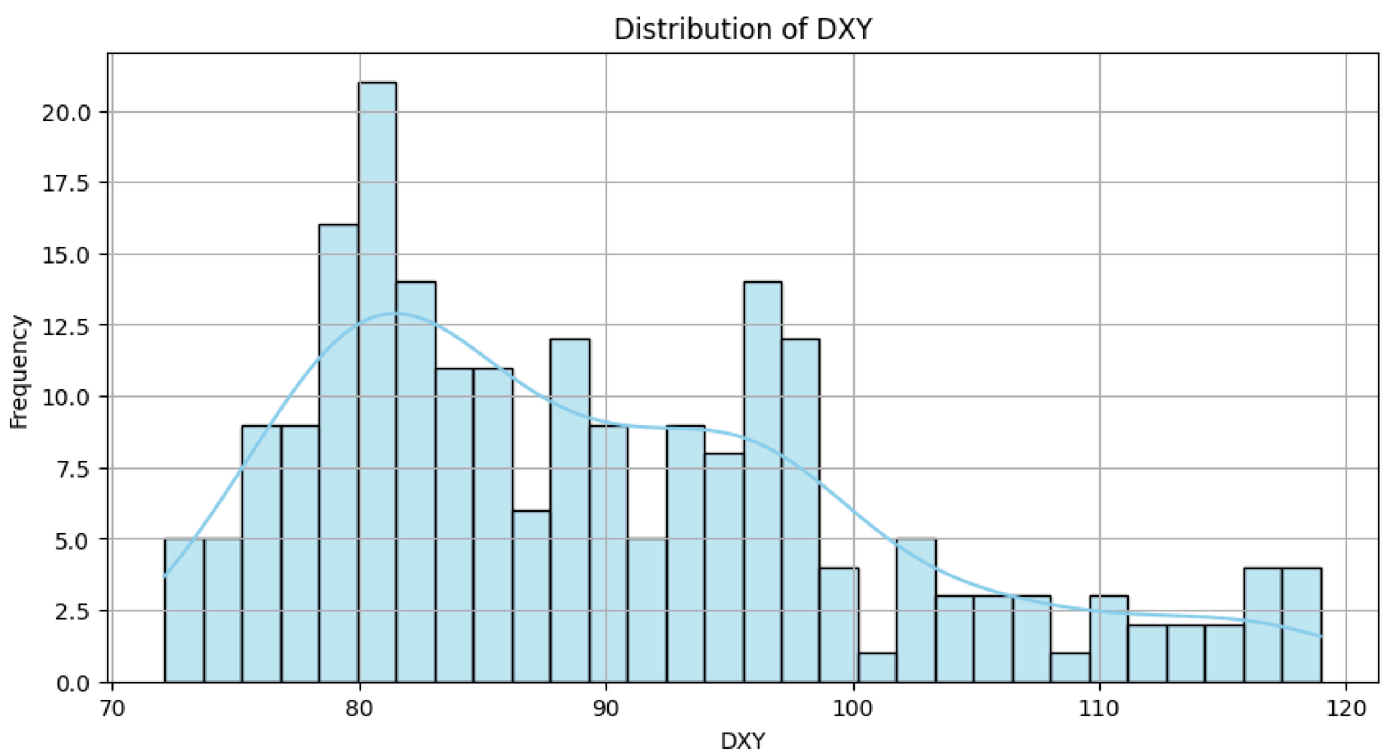

Distribution of WTI
Frequency
0
2
4
6
8
10
12
14
16
20
40
60
80
100
120
WTI

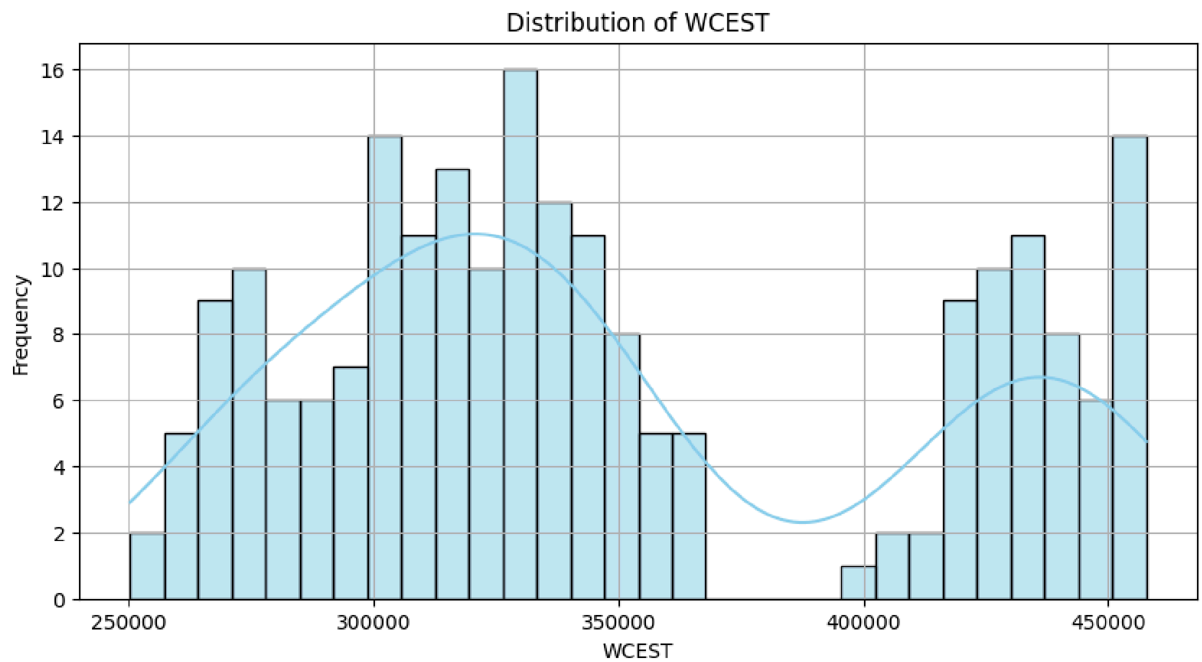

Distribution of CPI
Frequency
0
5
10
15
20
25
180
200
220
240
260
280
300
CPI


Distribution of GDP
Frequency
GDP
30
25
20
15
10
5
0
−2
0
2
4
6

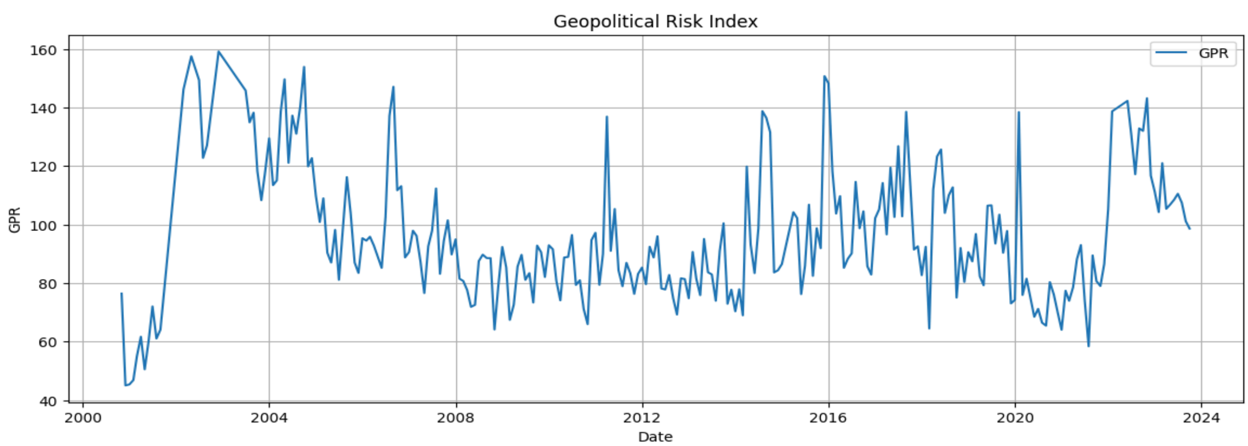

Distribution of RUR
Frequency
RUR
20
15
10
5
0
50
100
150
200
250
300
350
400

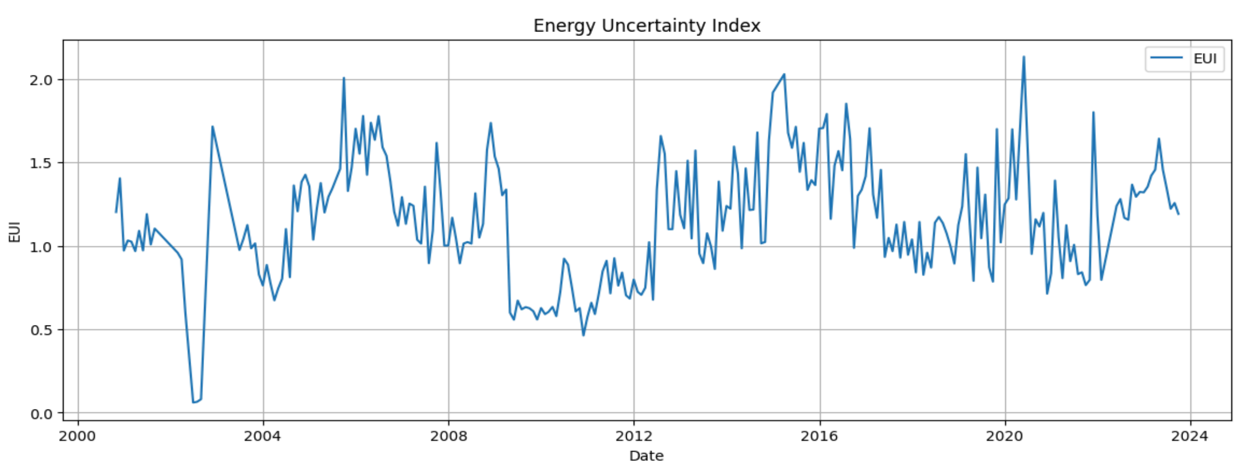

Distribution of VIX
Frequency
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
10
15
20
25
30
35
VIX

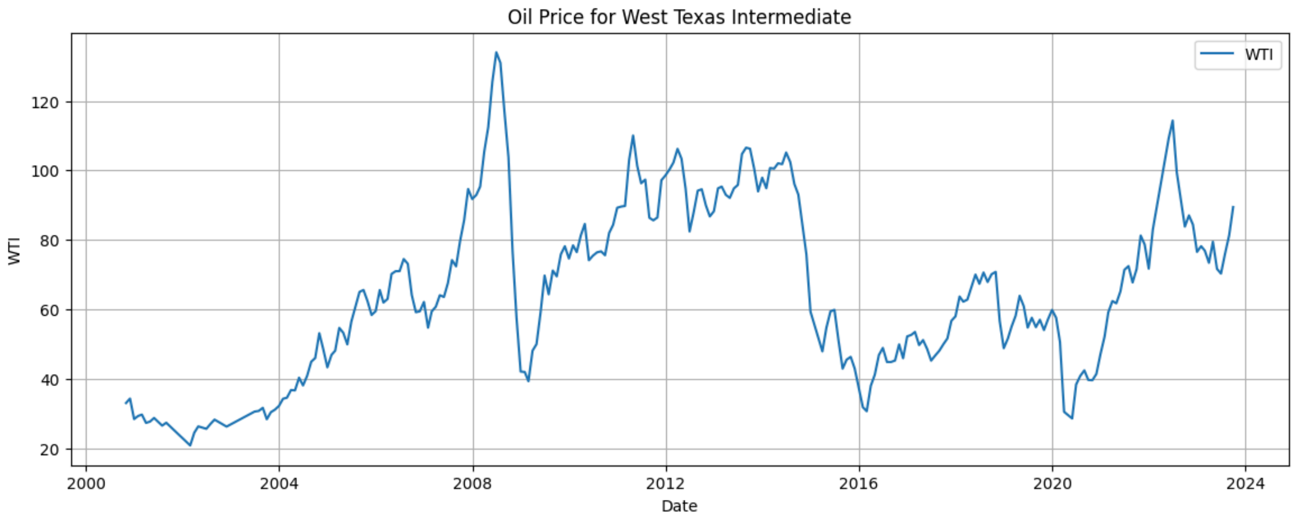

Distribution of DXY
Frequency
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
70
80
90
100
110
120
DXY

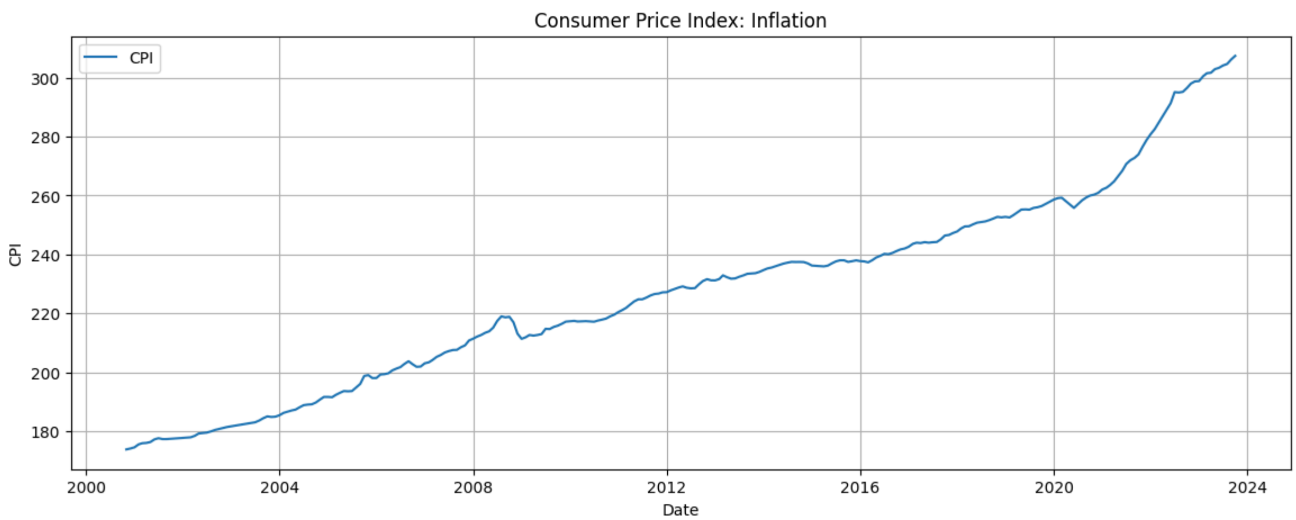


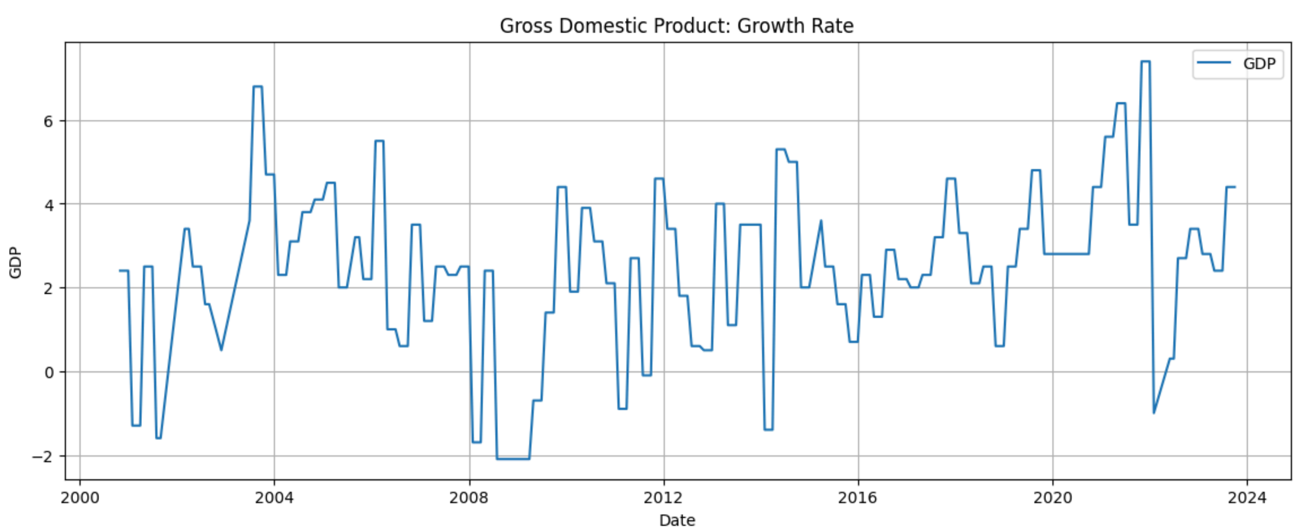
Time Series Plots

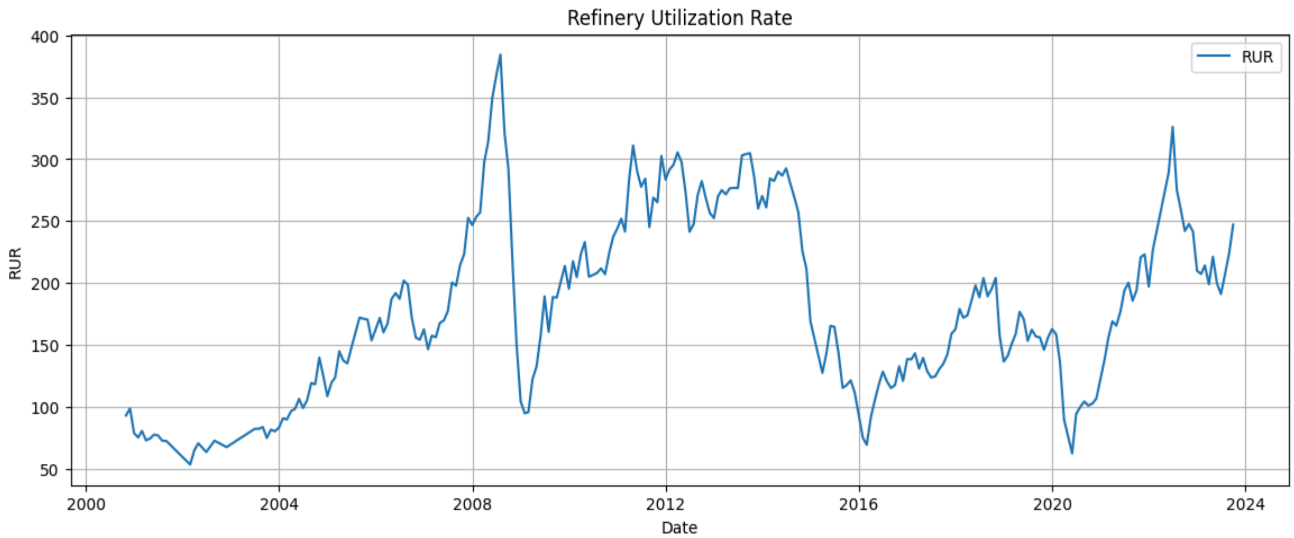


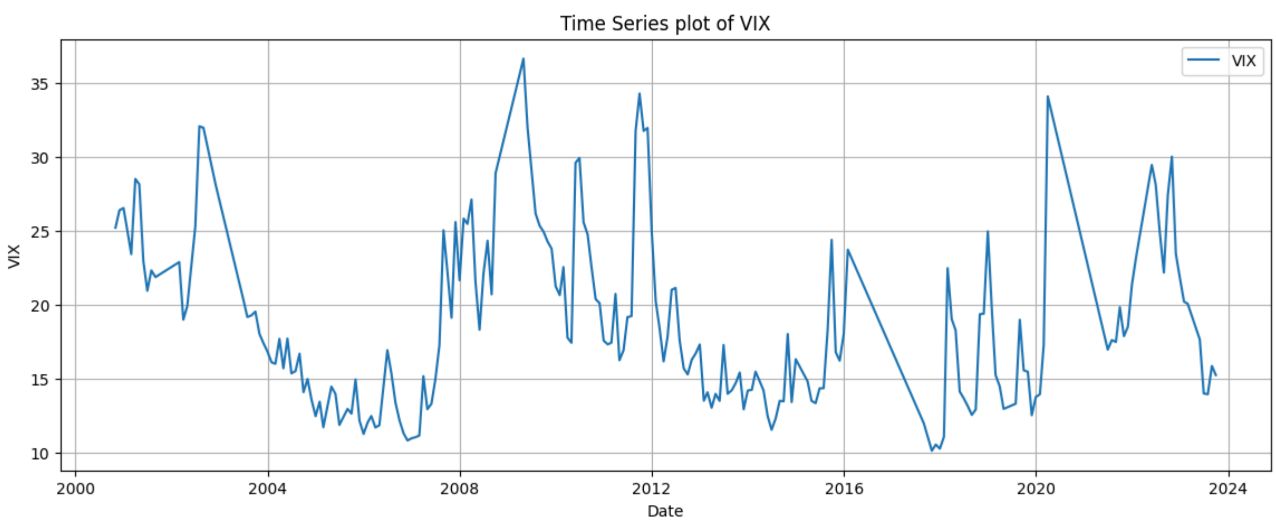


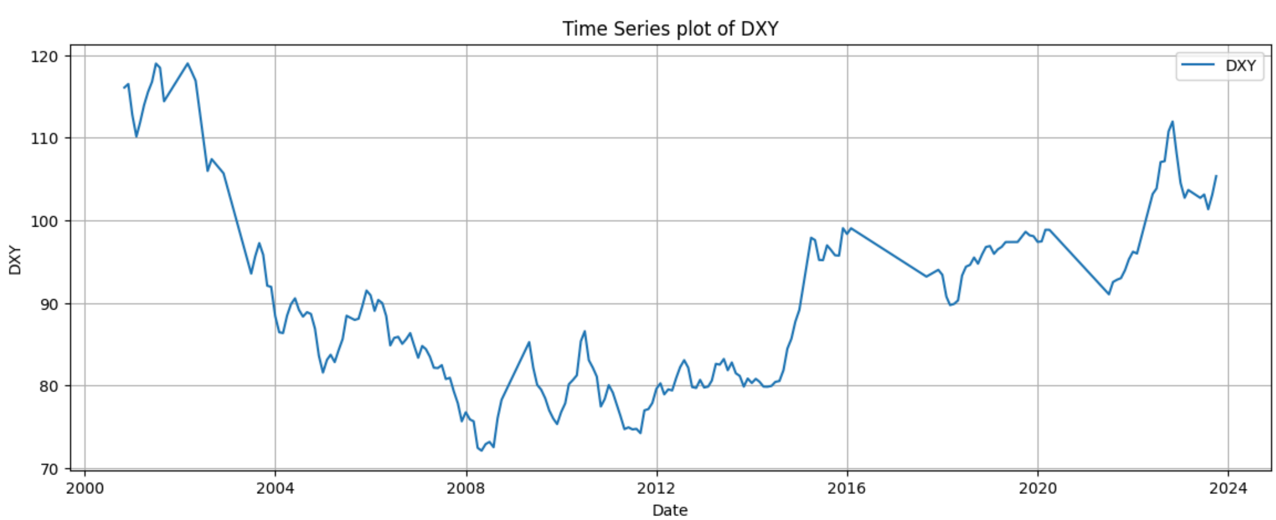

Oil Price for West Texas Intermediate
WTI
120
100
80
60
40
20
2000
2004
2008
2012
2016
2020
2024
Date

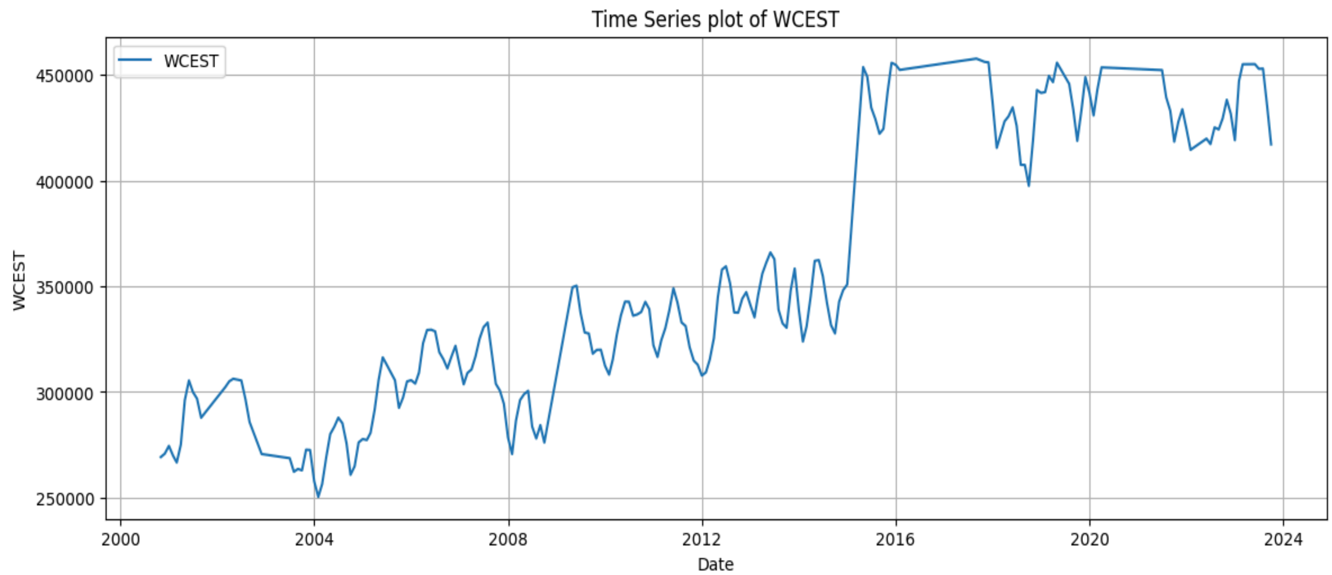

Consumer Price Index: Inflation
CPI
300
280
260
240
220
200
180
2000
2004
2008
2012
2016
2020
2024
Date


Gross Domestic Product: Growth Rate
GDP
6
4
2
0
−2
2000
2004
2008
2012
2016
2020
2024
Date

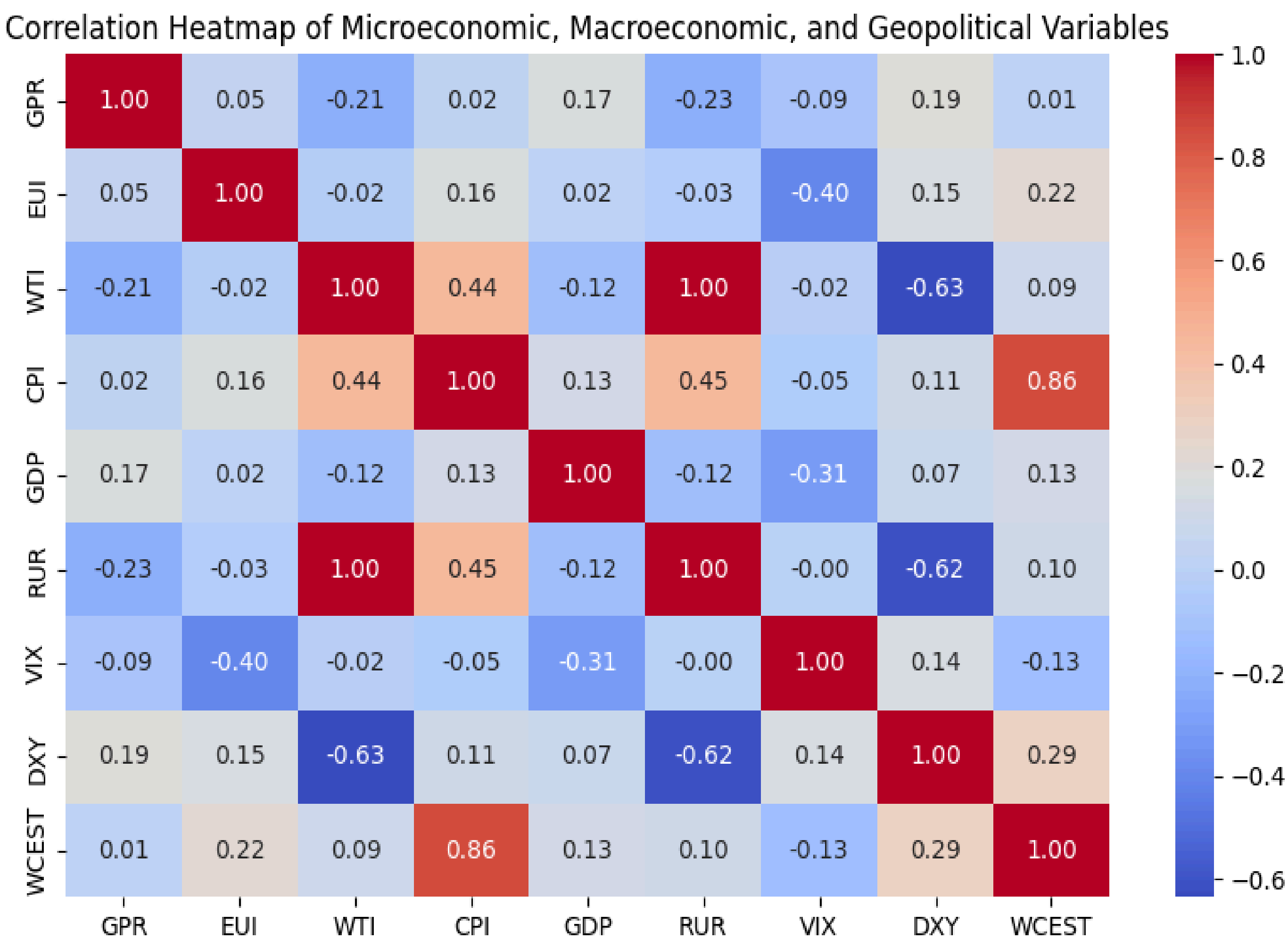

Refinery Utilization Rate
RUR
400
350
300
250
200
150
100
50
2000
2004
2008
2012
2016
2020
2024
Date

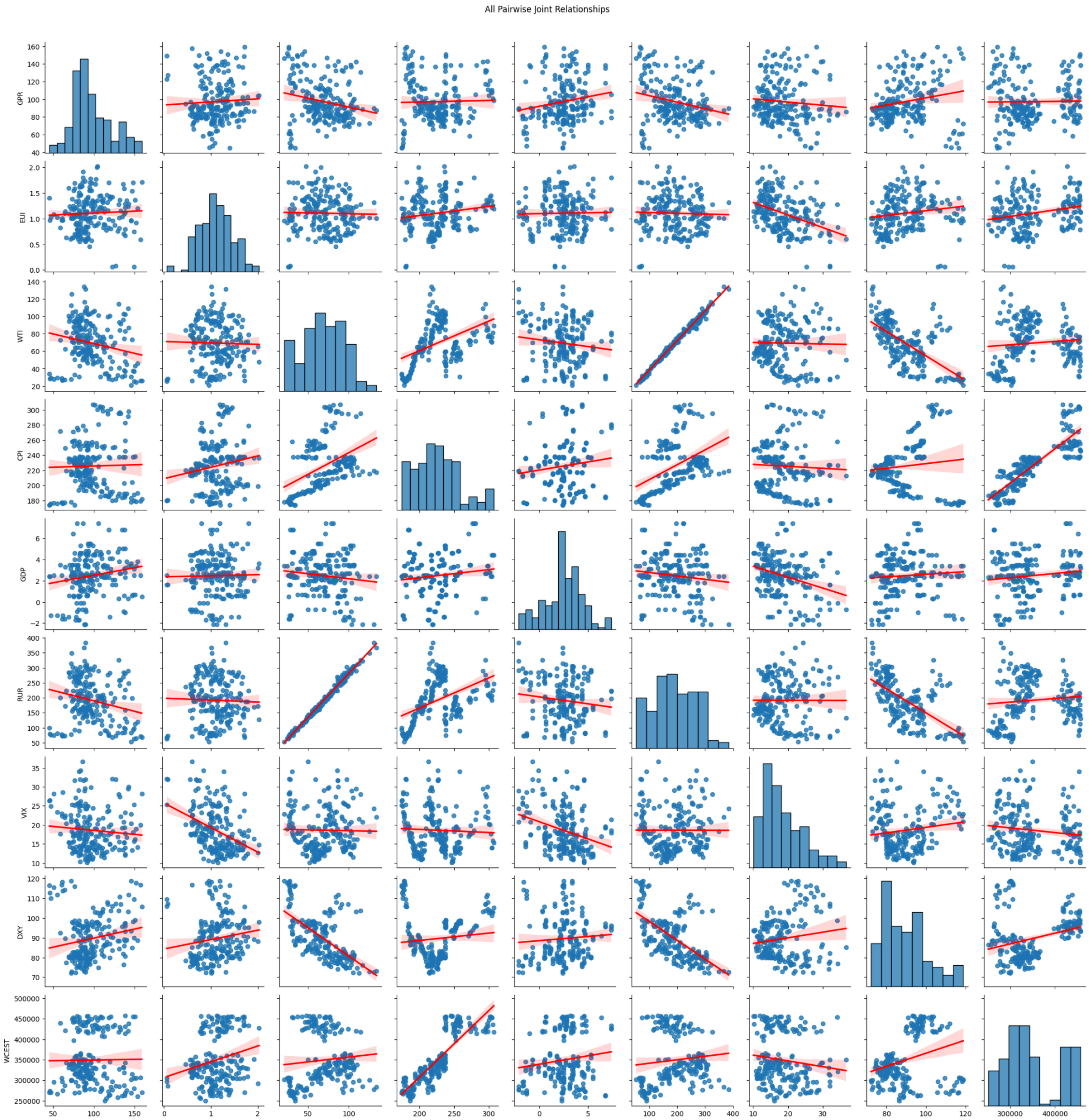

Time Series plot of VIX
VIX
35
30
25
20
15
10
2000
2004
2008
2012
2016
2020
2024
Date

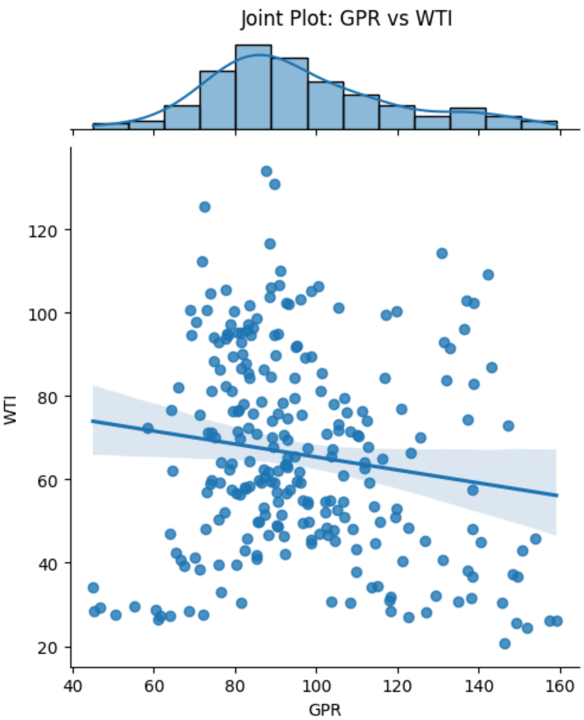

Time Series plot of DXY
DXY
120
110
100
90
80
70
2000
2004
2008
2012
2016
2020
2024
Date

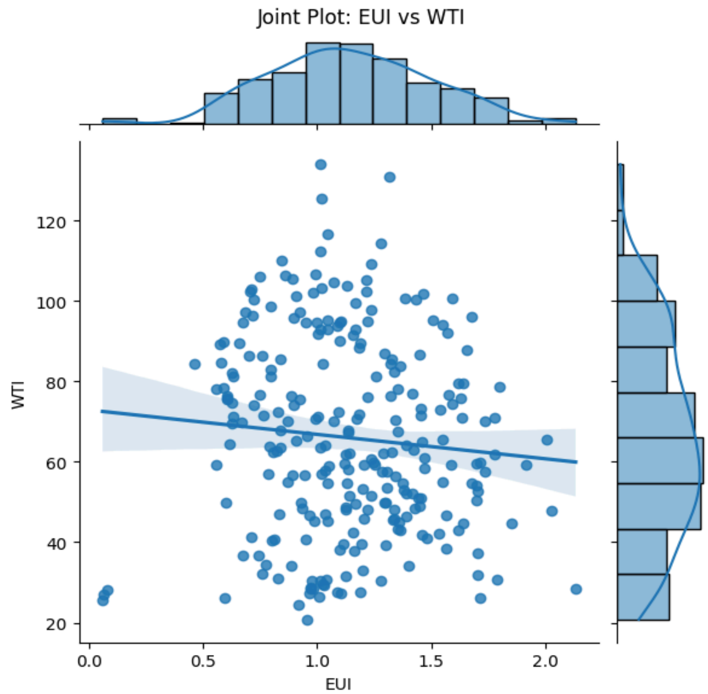


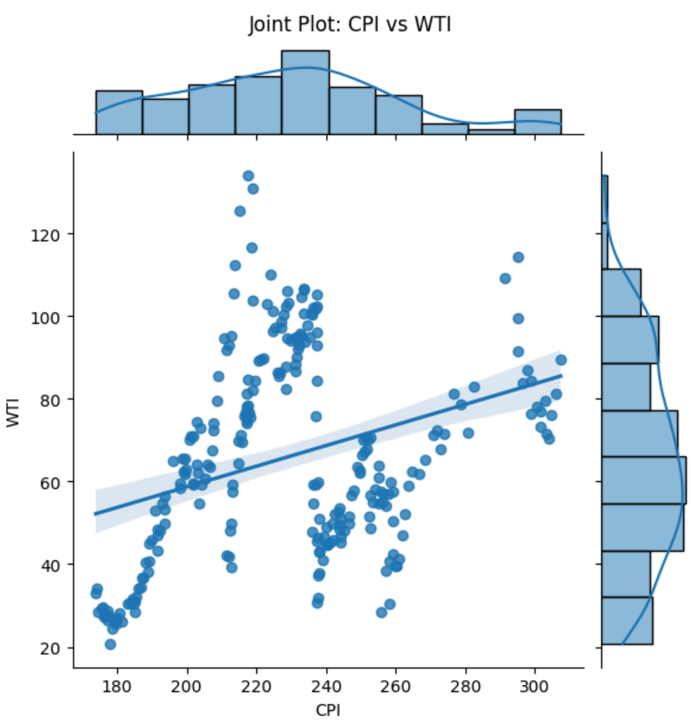
Multivariate Plots

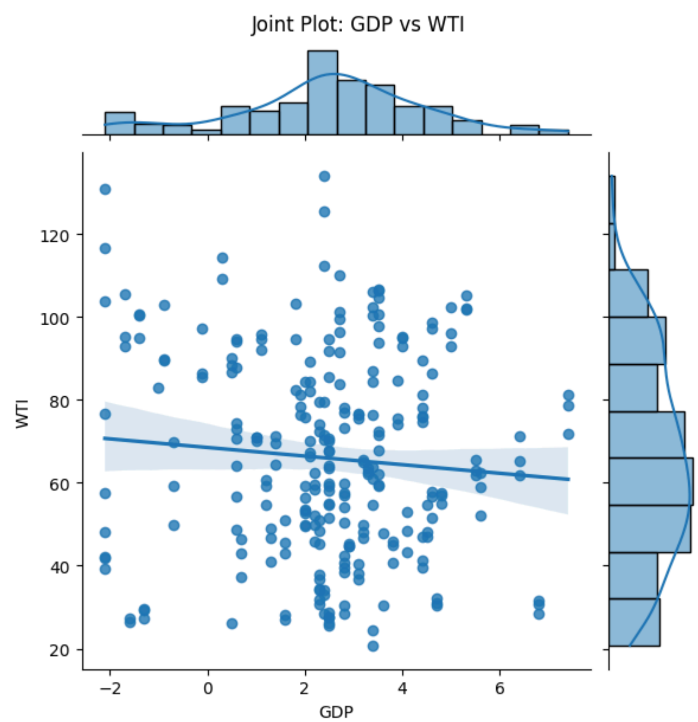


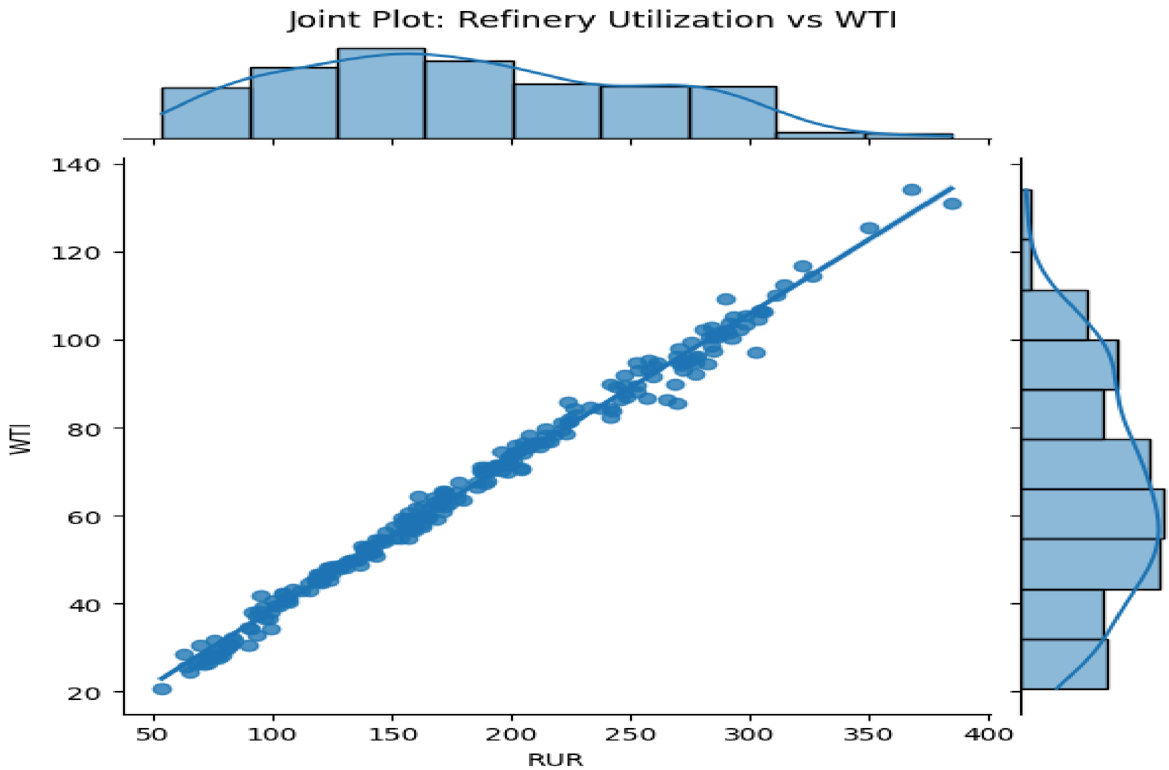

All Pairwise Joint Relationships
GPR
EUI
WTI
CPI
GDP
RUR
VIX
DXY
WCEST


Joint Plot: GPR vs WTI
WTI
GPR
20 40 60 80 100 120
40 60 80 100 120 140 160

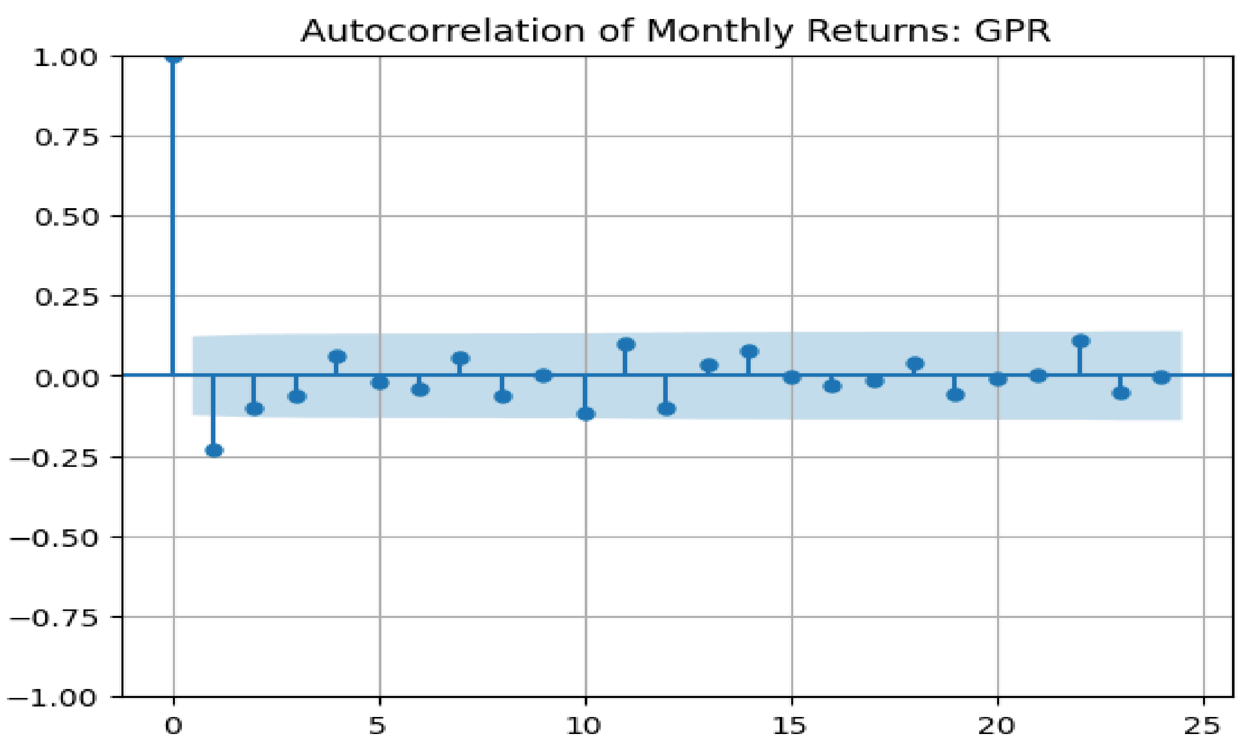

Joint Plot: EUI vs WTI
WTI
EUI
20 40 60 80 100 120
0.0 0.5 1.0 1.5 2.0

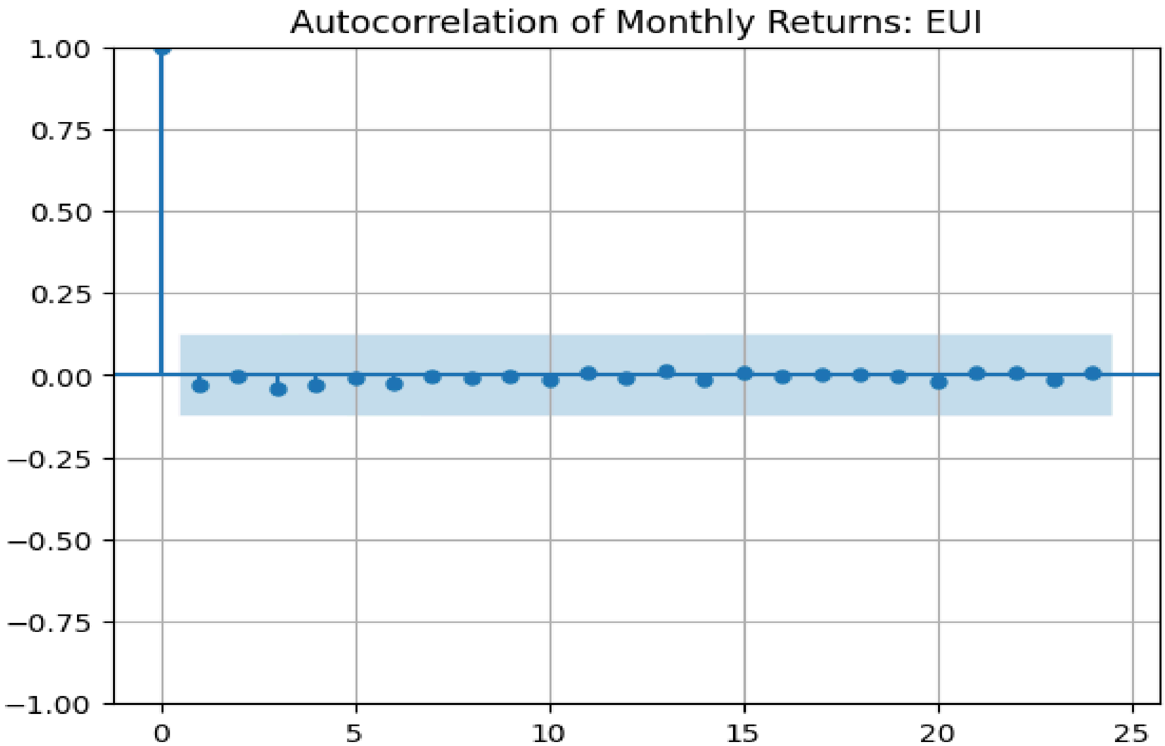

Joint Plot: CPI vs WTI
WTI
CPI
20 40 60 80 100 120
180 200 220 240 260 280 300

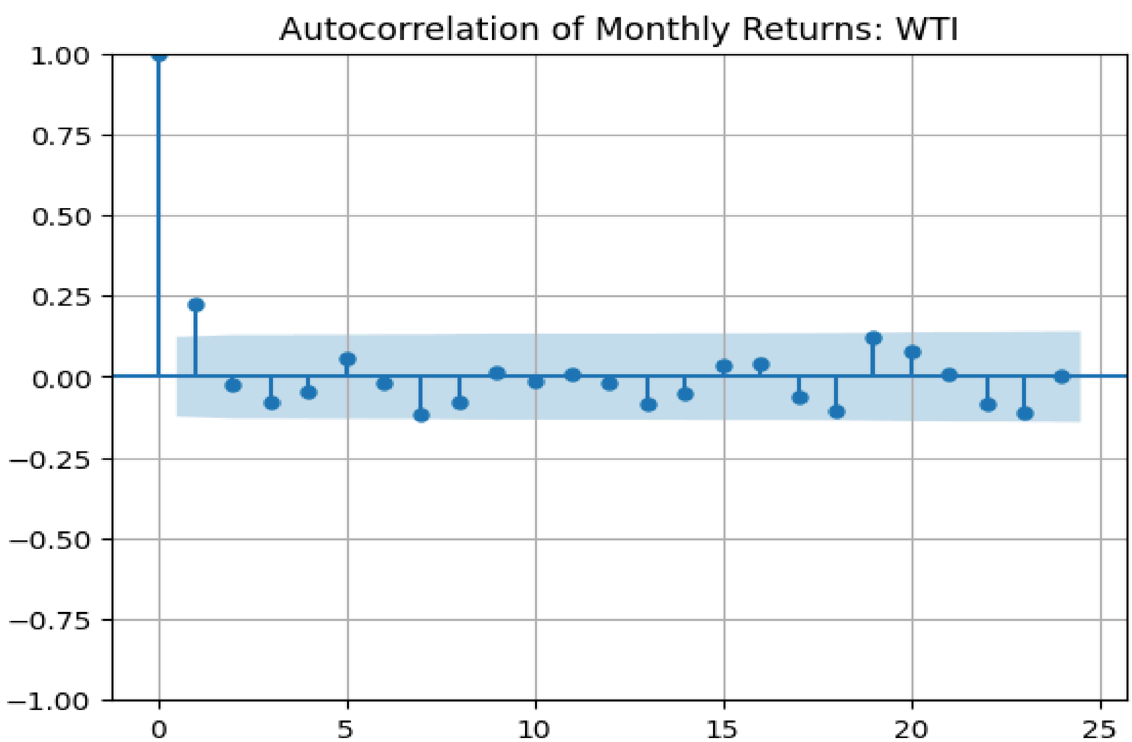

Joint Plot: GDP vs WTI
WTI
GDP
20 40 60 80 100 120
−2 0 2 4 6

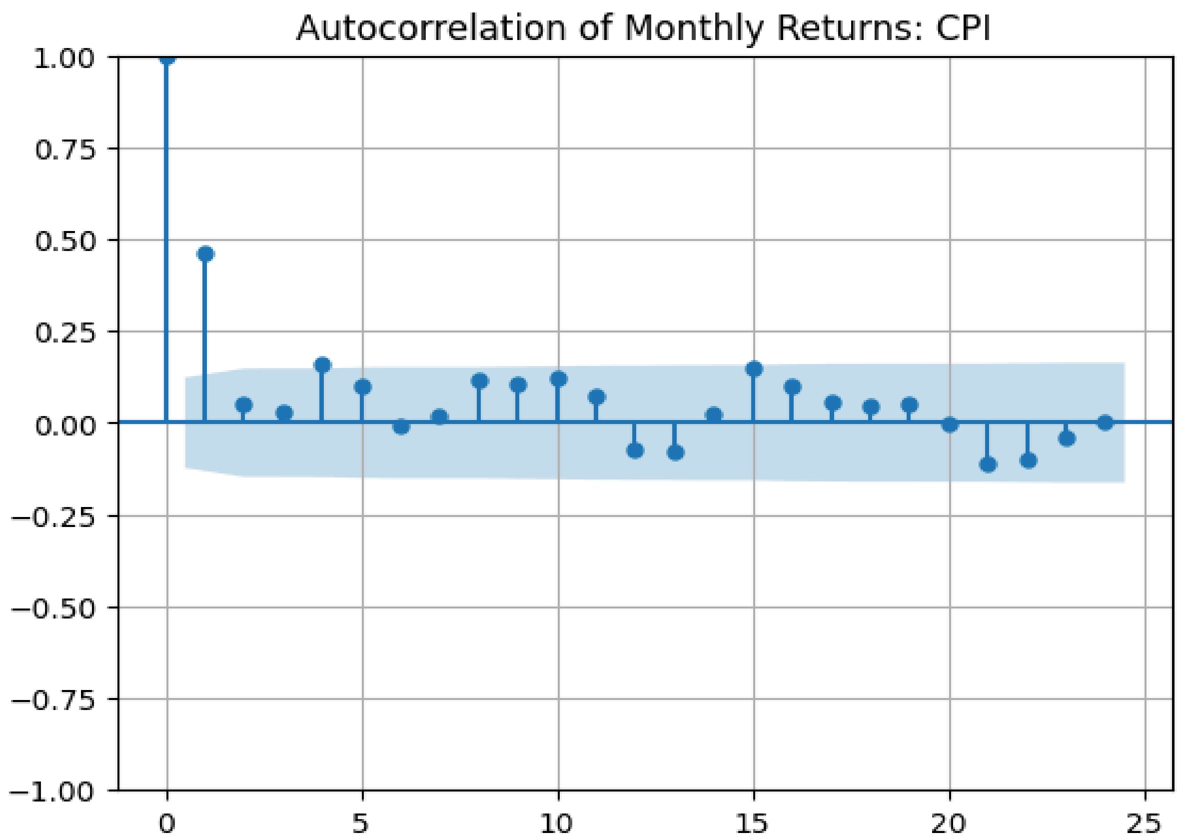


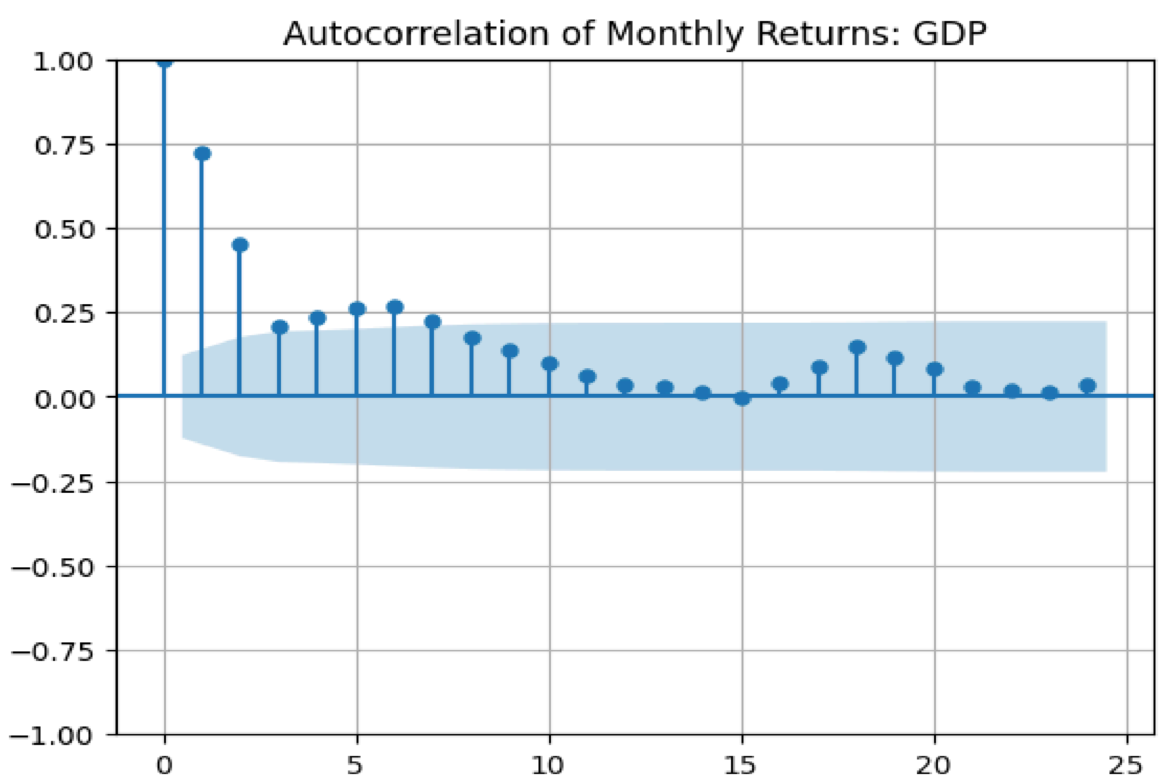
Autocorrelation (ACF) Plots

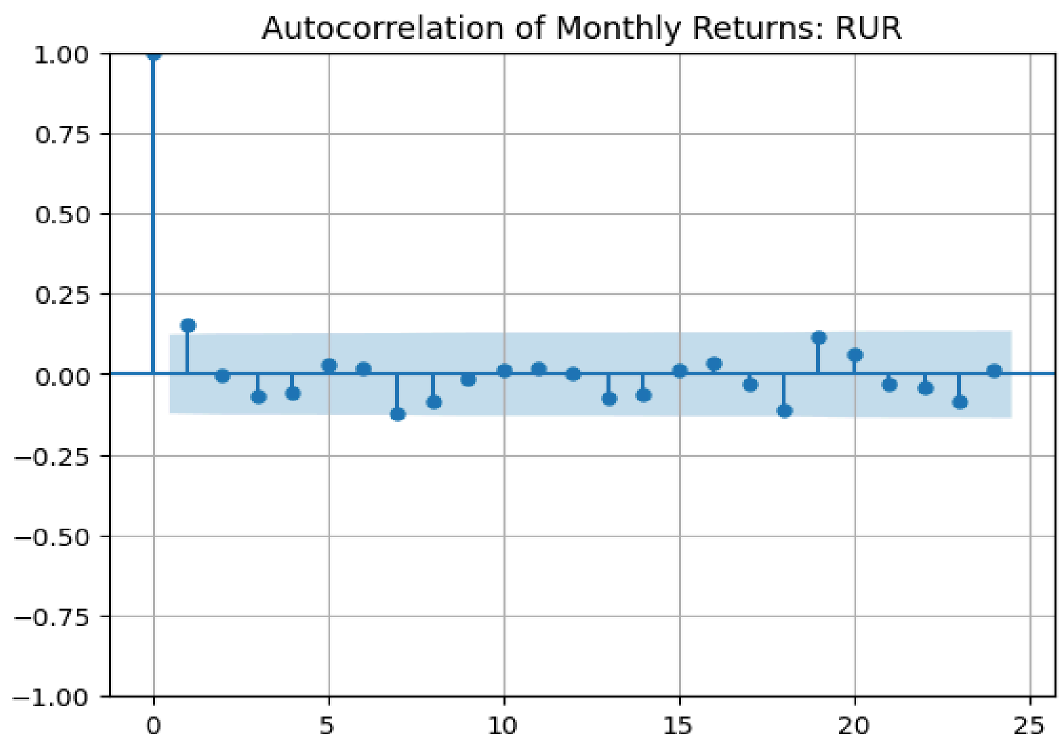


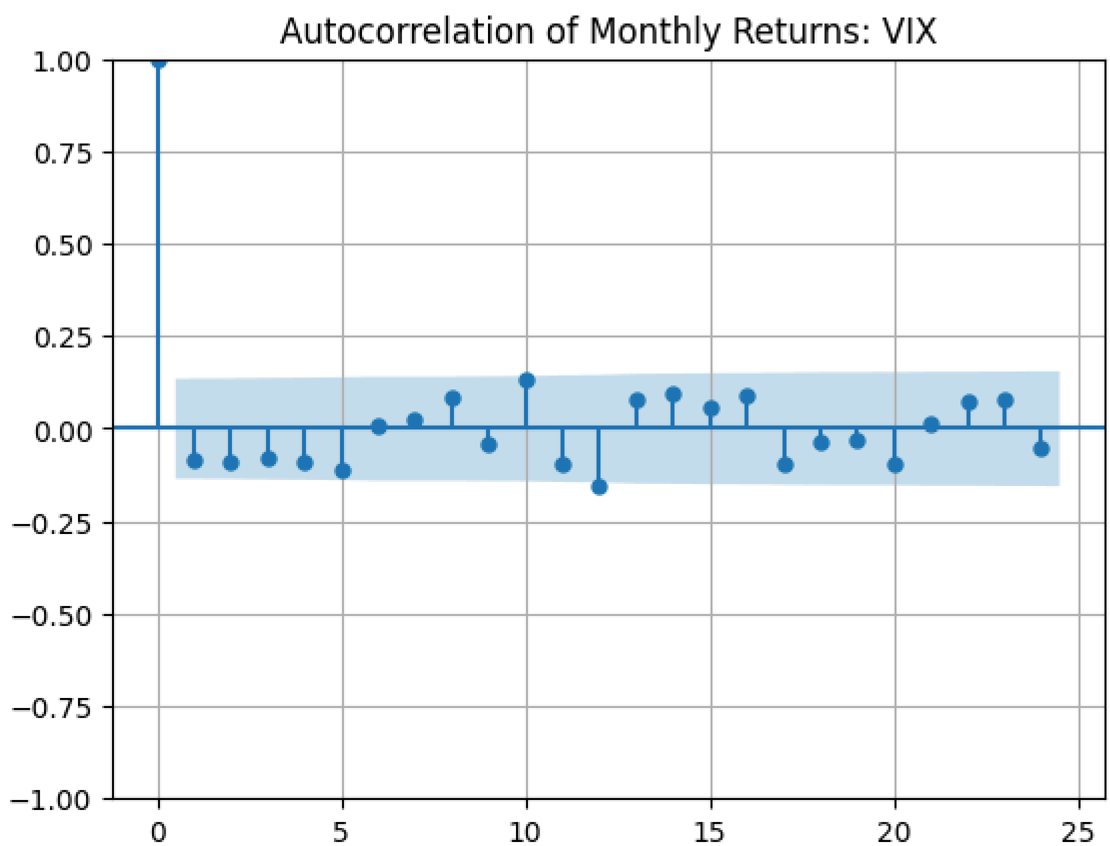

Autocorrelation of Monthly Returns: EUI
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25

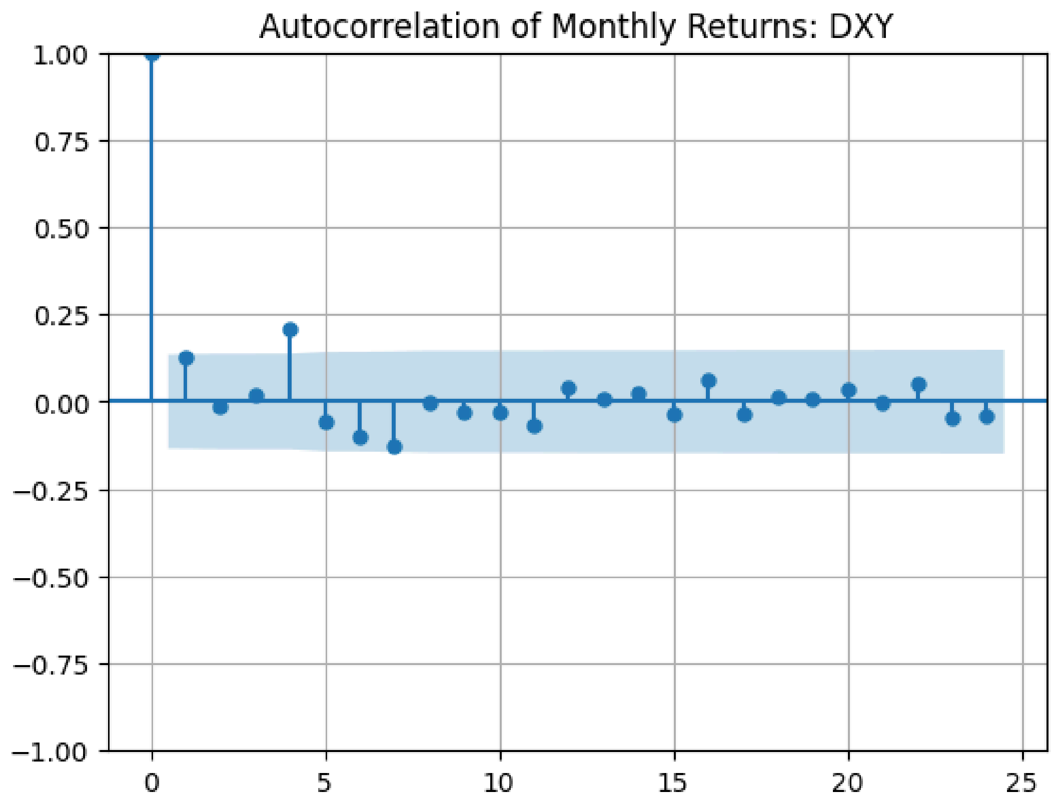

Autocorrelation of Monthly Returns: WTI
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25

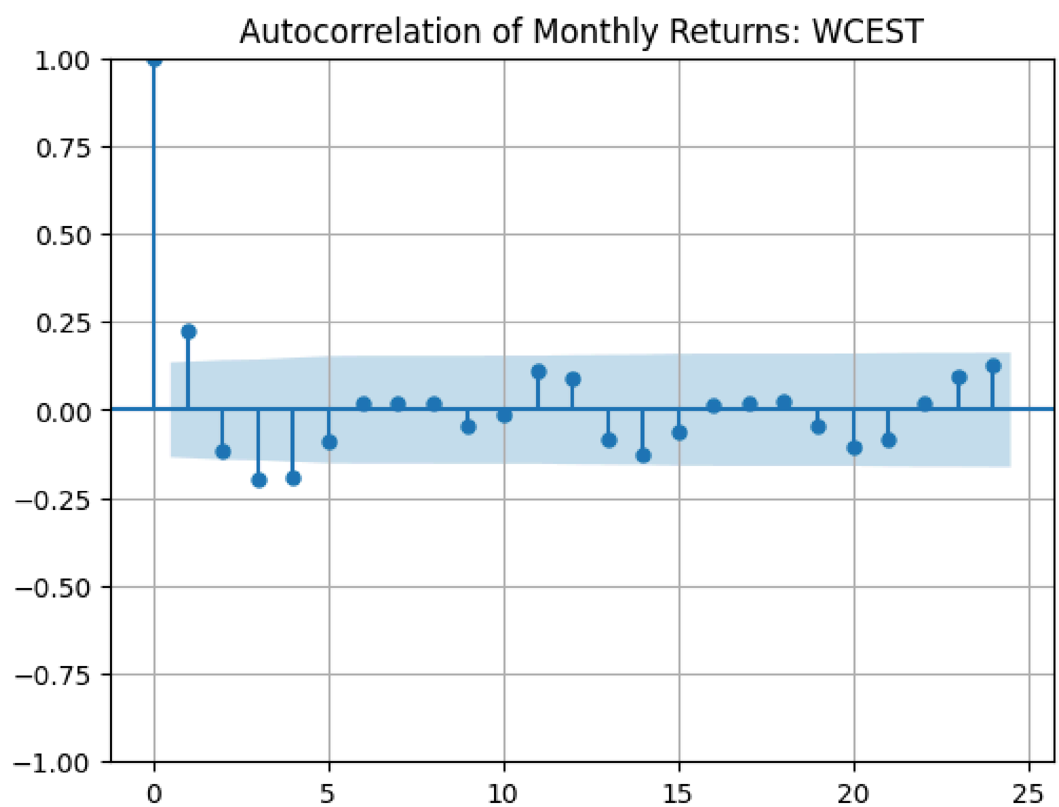

Autocorrelation of Monthly Returns: CPI
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25


Autocorrelation of Monthly Returns: GDP
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25


Autocorrelation of Monthly Returns: RUR
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25


Autocorrelation of Monthly Returns: VIX
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25


Autocorrelation of Monthly Returns: DXY
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25


Autocorrelation of Monthly Returns: WCEST
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25

Step 8 - Oil vs other asset prices

What makes oil prices look different from other asset prices?

- The distribution plots of WTI reveal heavy tails, meaning extreme oil price changes are more common than under a normal distribution.
- The time series plot of WTI shows sharp spikes and crashes, reflecting sensitivity to geopolitical shocks, supply chain disruptions, or policy changes.
- Compared to typical asset prices, oil prices exhibit volatility clustering — periods of calm are followed by turbulence, often due to non-market factors like war, OPEC decisions, or natural disasters.
- There is also some seasonality visible in the macro series, especially CPI and utilization rates.

What types of distributions do oil returns have?

- WTI and other variables (e.g., GPR, EUI) exhibit non-normal, skewed distributions.
  The distribution plots show:
  - Fat tails in WTI returns (risk of large swings)
  - Right skewness in GPR — long periods of low risk interrupted by spikes.

- This suggests higher kurtosis, which invalidates normality assumptions and justifies using Bayesian methods that can handle such uncertainty.

What type of autocorrelation do the oil returns have?

- The autocorrelation plots (ACF) for WTI returns show:
  - Very low or no significant autocorrelation in raw returns, suggesting they resemble white noise.
  - However, squared returns (if plotted) might show volatility clustering — a key stylized fact in commodities.

- Other variables like refinery utilization and GDP show some seasonal autocorrelation, which economic cycle effects could cause could cause.

Other stylized facts about oil prices:

- **Asymmetry in response:** Oil reacts more strongly to negative shocks (e.g., geopolitical tension) than positive signals.
- **Non-linear interactions:** Multivariate plots (e.g., GPR vs. WTI) show that relationships aren't strictly linear, reinforcing the need for a Bayesian Network rather than a simple regression model.
- High correlation between WTI and Refinery Utilization, moderate correlation with GDP and CPI, and weak correlation with GPR and EUI suggest that both economic and political forces shape oil prices.
- Lag effects likely exist (e.g., policy shocks influencing prices with a delay), which will be explored further in causal modeling.

Step 9 - Model summaries

Probabilistic Graphical Models

Probabilistic Graphical Models (PGMs) are a framework for representing complex distributions over many variables using graphs. They allow us to model and reason about uncertainty, which is central in financial decision-making. In finance, we often face questions like: What is the probability that oil prices will spike if geopolitical risk increases? Or, how does inflation impact asset returns? PGMs help us structure these relationships and perform inference efficiently.

PGMs combine the principles of probability theory and graph theory. The graphical structure makes dependencies explicit, allowing us to focus on relevant variables while ignoring irrelevant ones. This is critical in financial models where data is high-dimensional and noisy. There are two main types of PGMs:

1. **Belief Networks (Bayesian Networks):** These are directed acyclic graphs (DAGs) where nodes represent random variables and edges represent conditional dependencies. Each node comes with a conditional probability distribution (CPD) given its parents. Bayesian Networks are particularly useful for modeling causal relationships and uncertainty. In the context of oil pricing, a Bayesian Network can model how factors like geopolitical risk (GPR), energy uncertainty (EUI), inflation (CPI), and GDP interact to influence oil prices. For example, if GPR rises, it might increase oil prices directly or indirectly via EUI. Bayesian networks allow us to trace these paths of influence.

2. **Markov Networks (Markov Random Fields):** Markov Networks are undirected graphs. They are better suited for situations where we model symmetric relationships without a natural direction of causality. Each clique (fully connected subset of nodes) has a potential function, and the joint distribution is represented as a product of these potentials.

Markov Networks are often used in portfolio optimization problems, where assets are related through correlations rather than directional influences. For instance, we might model the joint behavior of a set of commodity prices or the co-movement of interest rates.

We use PGMs in financial engineering due to their scalability, interpretability, and flexibility. PGMs can handle large sets of variables, which is common in macroeconomic and financial systems. The graph makes it easier to understand dependencies. They support both causal inference and prediction. In our project, we rely on Bayesian Networks because they support causal modeling, which is key to understanding what drives oil price changes.

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Parameter Learning vs Structure Learning

Once a PGM structure is chosen, we must determine how to make it operational. This involves two key tasks: structure learning and parameter learning. In financial modeling, these steps allow us to make our models both data-driven and interpretable.

Parameter Learning is the process of estimating the probabilities or parameters associated with the graphical model, assuming the structure is already known. In a Bayesian Network, this means evaluating the conditional probability distributions (CPDs) for each node given its parents. For example, if we model that the oil price depends on GPR and EUI, parameter learning would involve estimating the likelihood of different oil prices given various combinations of GPR and EUI levels. In practice, this is done using historical data and methods like Maximum Likelihood Estimation (MLE) or Bayesian Estimation.

Structure Learning, on the other hand, involves discovering the network structure itself, which variables are connected and in what direction. This is more challenging but crucial in high-dimensional data environments like finance. There are two main approaches:

Score-based methods: Try different structures and score them using criteria like BIC or AIC.
Constraint-based methods: Use statistical tests to infer dependencies.

Structure learning helps discover whether GDP directly influences oil prices or if its effect is mediated through inflation. This step is critical in preventing us from hardcoding assumptions and instead letting the data reveal relationships.

Financial systems are dynamic and interconnected. Understanding structure helps identify which indicators matter most. Parameter learning ensures accurate forecasts and risk assessments. Together, they enable us to build data-driven yet interpretable models. Ultimately, good structure learning reveals the causal web of macroeconomic and geopolitical influences, while parameter learning calibrates our model to historical reality.

Markov Chains and Markov Blankets

In probabilistic modeling, Markov Chains and Markov Blankets are essential tools for simplifying complex dependencies and understanding temporal or local behavior.

Markov Chains are a type of stochastic process where the probability of transitioning to the next state depends only on the current state, not the full history. This property is known as memoryless. This "memoryless" property makes them powerful for modeling regime changes in finance. For example, in oil markets, we might model states like "stable prices," "bull run," and "crash." A Markov Chain can estimate the probability of moving from stability to a crash, which is useful for risk management and scenario planning. Each state has transition probabilities that define the dynamics of the system.

Markov Chains are often used in Credit rating migrations (e.g., AAA to AA), Volatility regime switching (low to high volatility), and Sequential modeling of macroeconomic conditions.

Markov Blankets, on the other hand, are a concept from Bayesian Networks. The Markov Blanket of a node is the minimal set of nodes that, once known, renders the node conditionally independent of the rest of the network. Specifically, a node's Markov Blanket includes its parents, its children, and other parents of its children. This is especially useful in large financial networks. For instance, to predict the WTI oil price, the Markov Blanket might include GPR, refinery utilization, and inflation. Once we condition on these, we can ignore the rest of the network.

- Markov Chains simplify temporal prediction and regime modeling.
- Markov Blankets isolate local dependencies, improving inference efficiency.

In our crude oil pricing model, Markov Chains help track transitions between market states, while Markov Blankets help localize the variables most relevant to price prediction. Both concepts reduce noise, increase interpretability, and make the Bayesian Network more efficient and actionable.

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Step 10 - Inferred Causality

Algorithm 1- Pseudocode for Inferred Causality

The pseudocode below outlines the logic behind identifying causal relationships within a Bayesian framework. The algorithm revolves around the three major points below.

- Structure learning forms the backbone.
- The Markov Blanket helps isolate relevant predictors
- The output is a provable, interpretable causal network

Input:

- **Data:** Cleaned and merged time series dataset (df_merged)
- **Target variable:** Oil Price (WTI)
- **Variables:** All other macroeconomic, microeconomic, and geopolitical indicators
- **MaxDepth:** Maximum depth for exploring dependencies (optional)

Output:

- **CausalGraph:** A directed graph showing inferred causal relationships
- **Markov Blanket (WTI):** Set of variables that directly influence WTI

Steps:

1. Preprocess data
    a. Normalize or standardize all variables
    b. Handle missing values

2. Initialize empty graph structure (CausalGraph)

3. Perform Structure Learning
    a. Use score-based or constraint-based method (e.g., PC Algorithm or Hill-Climbing)
    b. Evaluate candidate network structures using scoring function (e.g., BIC or AIC)
    c. Select the best structure based on the score

4. Extract Markov Blanket for target (WTI)
    a. For the WTI node in the Bayesian Network:
        I. Identify its parents (direct causes)
        ii. Identify its children (effects)
        iii. Identify other parents of its children (indirect influences)
    b. Combine these into Markov Blanket (WTI)

5. Build a directed CausalGraph using the selected structure

a. Retain only edges that are part of the Markov Blanket (WTI)

b. Optionally, assign edge weights based on conditional probability strengths

6. Output CausalGraph and MarkovBlanket(WTI)

End.

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